



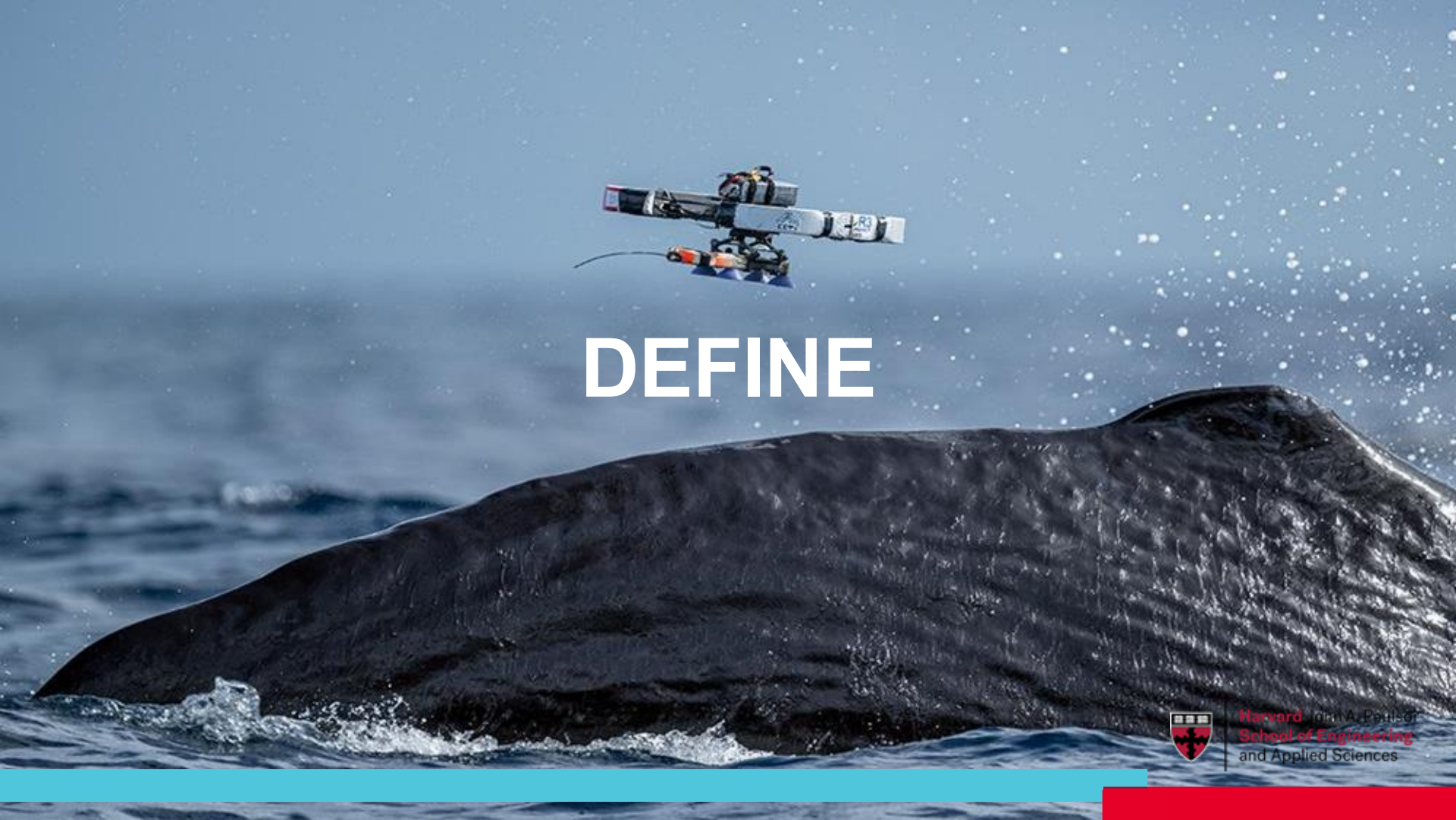
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Design, Fabrication and Test of a Modular Fixed-Wing VTOL Drone Platform for Sperm Whale Tracking and Localization

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Advisors: Stephanie Gil, Hammad Izhar

Thesis Presentation

A robotic underwater vehicle (ROV) is shown hovering in the air above a large whale as it breaches the ocean surface. The whale's dark, wet skin is visible, and water droplets are suspended in the air around it. The ROV has a white body with blue and orange accents and is equipped with various sensors and cameras.

DEFINE

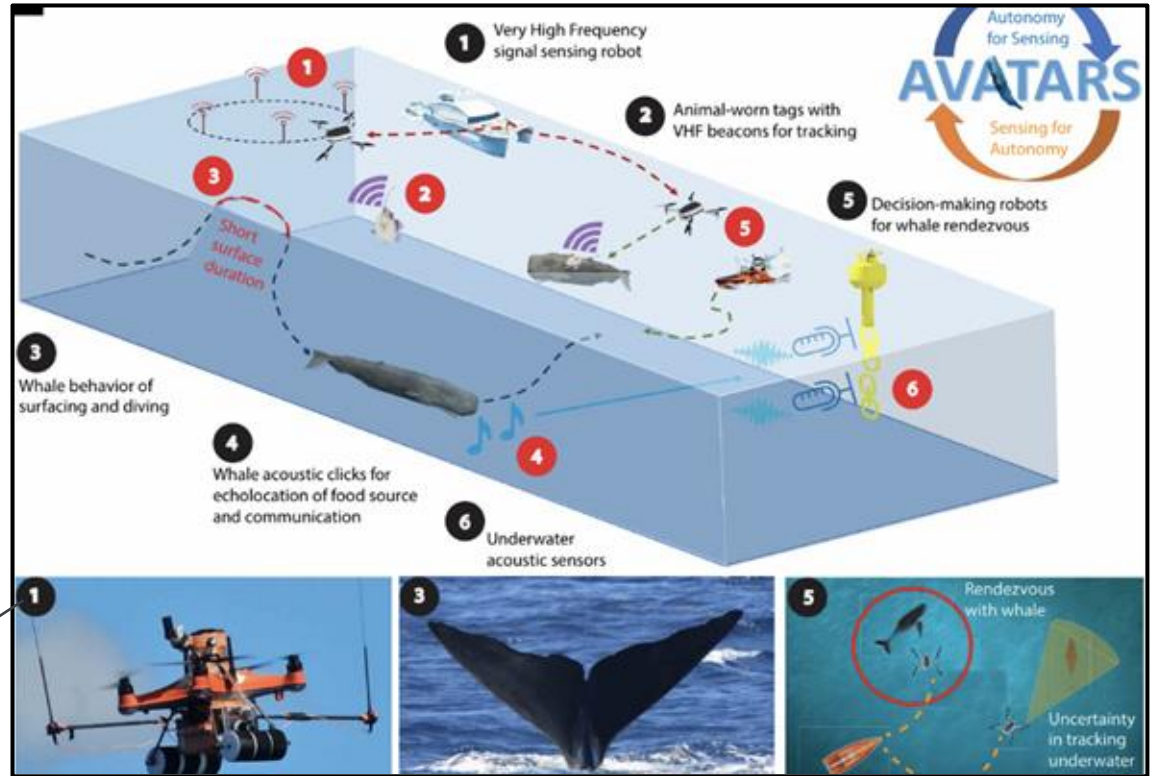


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Project CETI - Decoding Whale Communication

- Global initiative to decode sperm whale communication using machine learning & robotics
- Fieldwork in Dominica with whale-mounted VHF tags and acoustic sensors
- Requires autonomous, reliable tools for whale tracking and data collection in harsh marine environments

Project Ceti Current Quadcopter



Problem Statement

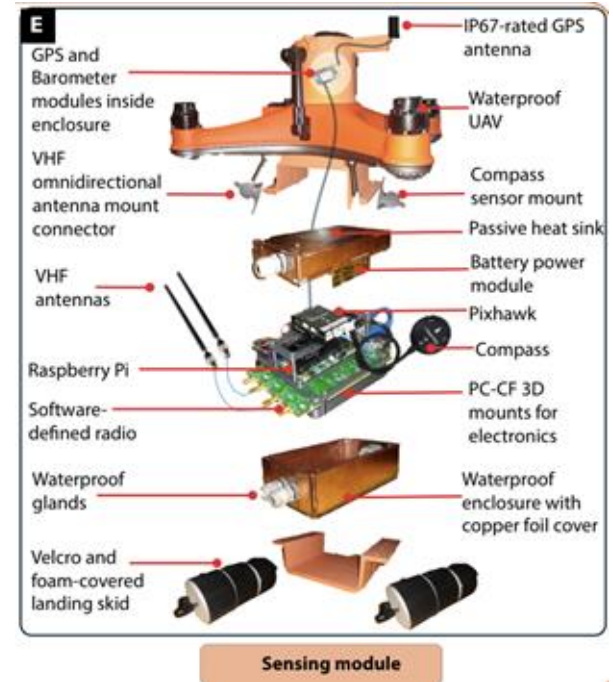


Current quadcopter drones are deployed from research vessels to locate and visually track VHF-tagged whales; however, they are not optimized for harsh marine conditions, have limited flight time, and struggle to provide sufficient antenna separation while remaining compact. This project addresses the problem of designing an aerial vehicle that can autonomously track VHF-tagged whales from a research vessel while being water-resistant, offering adequate flight endurance, staying compact for small-boat deployment, and providing reliable signal detection.

Existing Solutions

- **Current approach:** Side standoffs place VHF antennas ~1 m apart for isolation.
- **Drawbacks:** ↑ drag and bulk, higher snag/impact risk, less streamlined profile.
- **Quadcopter strengths:** Simple, VTOL, precise/robust control.
- **Key limitation:** Short endurance → reduced time-on-station and survey coverage.

Net effect: Achieves signal separation, but at the cost of efficiency and mission duration.



Technical Specifications

ID	Spec	Target	Verification
T1	Antenna Separation	1 m spacing	Distance measurement
T2	Weight	2 kg fuselage + payload, 3.5 kg total	Balance Measurement
T3	Size	$\leq 1\text{m} \times 1\text{m} \times 0.5\text{m}$	Dimension Inspection
T4	Durability	withstand 3.5 kg point load without permanent deformation	FEA (due to limited materials)
T5	Cost	$\leq \$1000$	BOM
T6	Endurance	≥ 45 min(1.5x to 2x current quadcopter max flight duration)	Hover/flight duration test
T7	Modularity	Critical components can easily detach for transportation (< 30 min assembly/configuration time)	Timed disassembly and reassembly

Proposed Solution



1



2



3



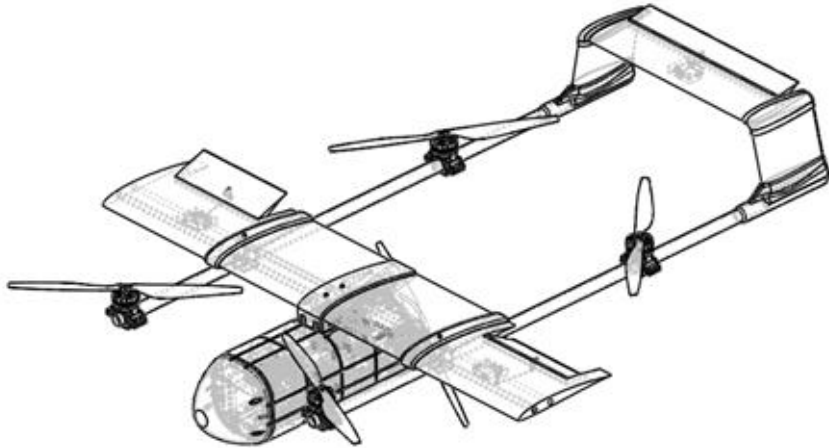
4



5

	1	2	3	4	5
Criteria	Quadcopter	Fixed-Wing	Tail-Sitter VTOL	Tiltrotor VTOL	Lift+Cruise VTOL (Selected)
VTOL Capability	Vertical takeoff/landing	Requires runway/launcher	Vertical takeoff/landing	Tilting rotors enable VTOL	Dedicated lift motors for vertical takeoff/landing
Hover Stability	Precise position hold	Cannot hover	Transition-sensitive hover	Active tilt stabilization	Stable hover using lift rotors
Endurance / Range	15-20 min (short range)	1-2 hr cruise	1 hr (mission dependent)	1-2 hr post-transition	1-2 hr cruise segment
Energy Efficiency	Rotor-dependent lift (power-intensive)	Aerodynamic lift efficiency	Efficient cruise; VTOL penalty	Efficient cruise; hover penalty	Efficient cruise; VTOL used only for transitions
Mechanical Complexity	Simple frame, few parts	Fixed structure	No tilt but complex transition	Servo tilt mechanisms and linkages	Separate lift and cruise propulsion
Control System Complexity	Standard PX4 control	Conventional forward flight	Requires mode-switch algorithms	Multi-axis coordination required	PX4 VTOL stack; separate flight modes
Payload Capacity	1-2.5 kg	2-3 kg (airframe dependent)	Limited by compact size	Scalable with motor power	2-3 kg with added lift margin
Launch / Recovery	Deck takeoff/landing	Requires open space/runway	Tail-down launch and landing	Precise tilt landing	Deck vertical takeoff/landing
Marine Suitability	Boat-deck compatible	Difficult recovery on moving deck	Feasible but delicate landings	Sensitive to turbulence near water	Designed for deck-based operation

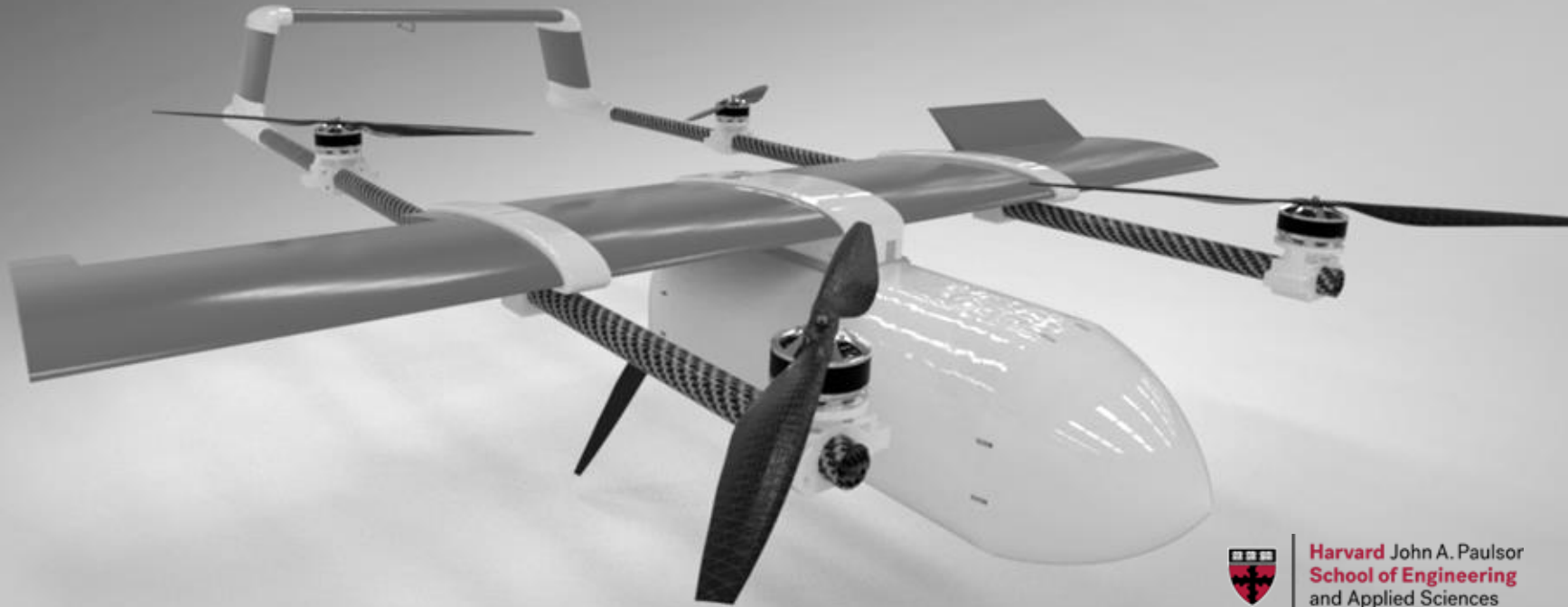
Proposed Solution



Lift + Cruise VTOL

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DESIGN



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Component Selection

2.1 VTOL Equations

$$F_{thrust} = F_{drag} + mg \quad (1)$$

$$= \frac{1}{2} \rho \omega^2 S C_{n\perp} + mg \quad (2)$$

$$v_i = \sqrt{\frac{T}{2\rho A_{tot}}} \quad (3)$$

$$A_{tot} = \sum A_{disk} \quad (4)$$

$$F_{thrust \text{ required}} = 38N \quad (5)$$



2.2 Fixed Wing Flight

The governing equation of flight is the Lift Equation:

$$L = \frac{1}{2} C_L \rho S v^2 \quad (9)$$

$$L = W = mg \quad (10)$$

$$v_s = \sqrt{\frac{2W}{\rho S C_L}} \quad (11)$$

$$v_s = 15.16[m/s] \quad (12)$$

$$v_{cruise} \approx 1.3 \cdot v_s \quad (13)$$

$$v_{cruise} = 19.7[m/s] \quad (14)$$

$$V_{ideal, \text{ fixed wing motor}} = \text{Pitch} \left(\frac{\text{in}}{\text{rotation}} \right) \cdot \text{RPM} \cdot \frac{1m}{39.3701in} \cdot \frac{1min}{60sec} = 24.6m/s$$



Component Selection

2.2 Fixed Wing Flight

$$P_{flight} = D(V) \cdot V_{cruise} \quad (15)$$

$$D(V) = \frac{1}{2} \rho v^2 S C_D \quad (16)$$

$$C_D = C_{D0} + k(C_L)^2 \quad (17)$$

$$k = \frac{1}{\pi e A R} \quad (19)$$

$$P_{drag} = \frac{1}{2} \rho v_{cruise}^3 S C_D = 91.9 \text{ Watts} \quad (20)$$

$$P_{flight} = \frac{P_{drag}}{\eta_{propulsive}} \approx 120 \text{ Watts} \quad (21)$$



6.3 Energy Calculations

$$E_{mission} = \sum E_i \quad (22)$$

$$E_i = P_i \times t_i \quad (23)$$

$t_{takeoff} + t_{landing}$ is around 5 minutes meaning that:

$$E_{hover} = 34.26 \text{ Wh} \quad (27)$$

$$E_{flight} = P_{flight} \cdot t_{flight} \quad (28)$$

Assuming we want atleast 45 minutes of flight:

$$E_{flight} = 90 \text{ Wh} \quad (29)$$

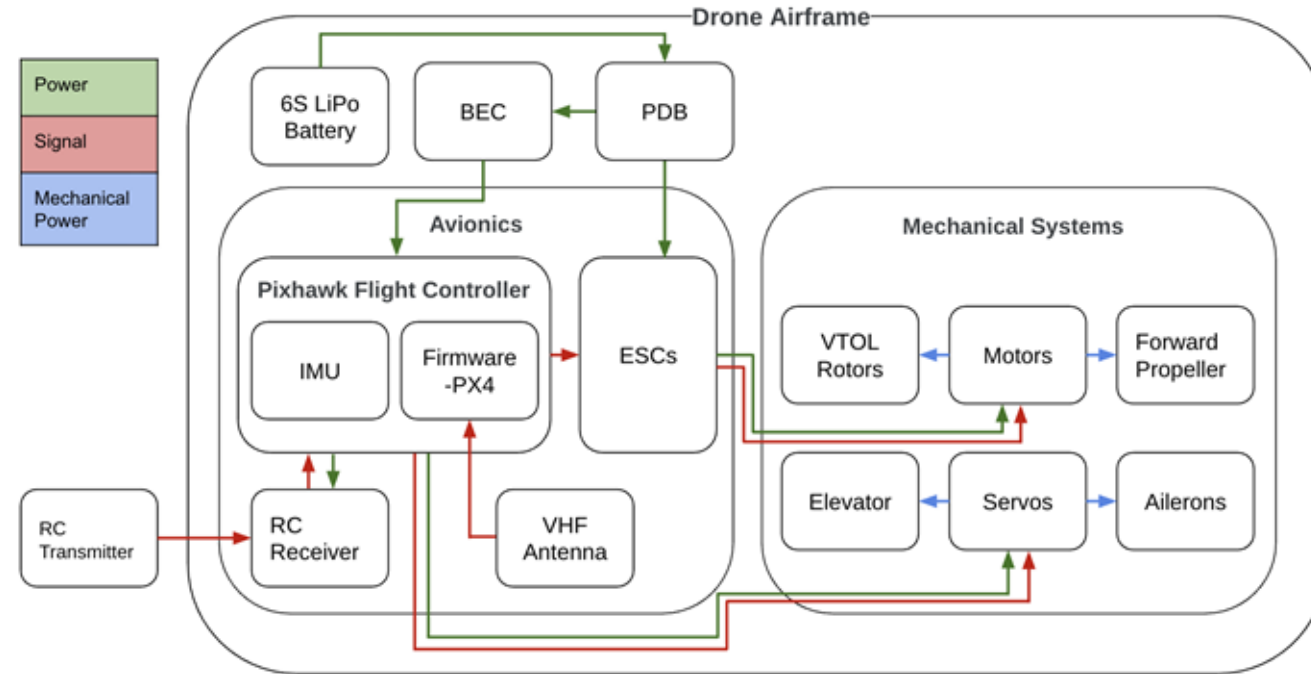
$$E_{mission} = E_{hover} + E_{flight} = 124.26 \text{ Wh} \quad (30)$$

$$C_{battery} = \frac{E_{mission}}{V_{battery}} \quad (31)$$

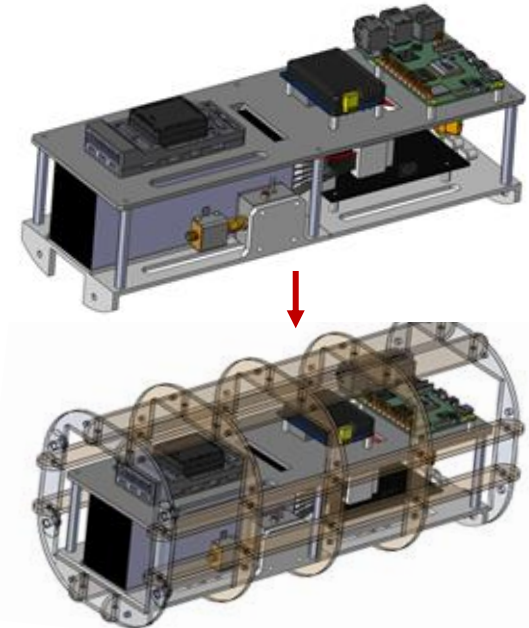
A standard 6s drone battery operates at around 22.2 Volts meaning that:

$$C_{battery} = \frac{124.26 \text{ Wh}}{22.2 \text{ V}} \approx 5.56 \text{ Ah} \approx 6 \text{ Ah} = 6000 \text{ mAh} \quad (32)$$

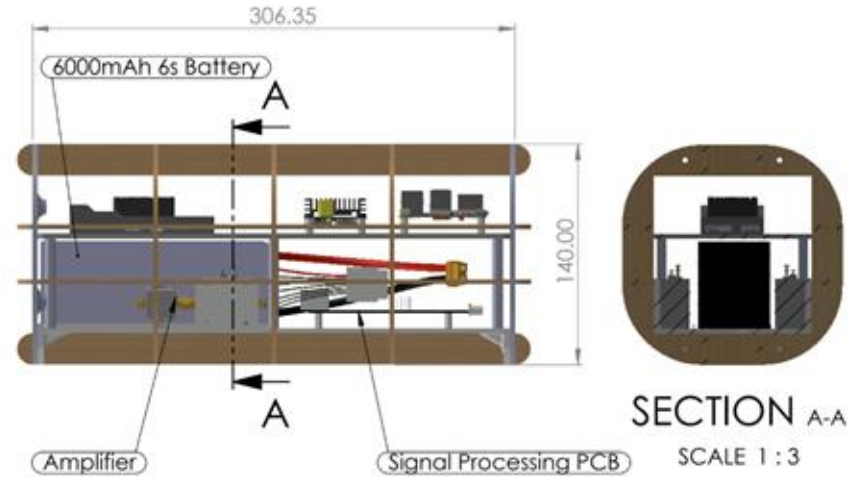
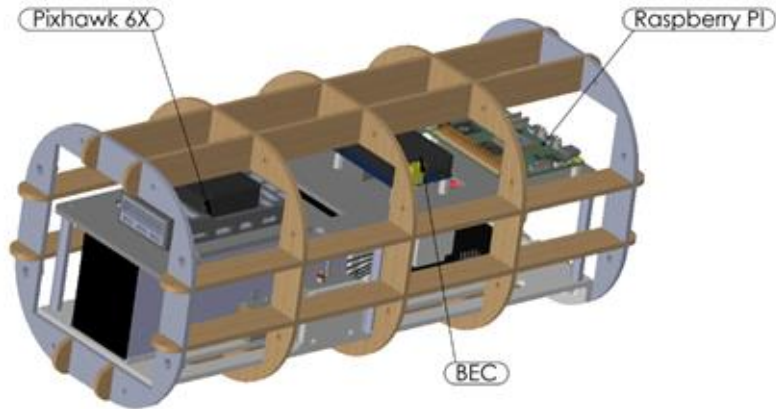
Electronic System Diagram



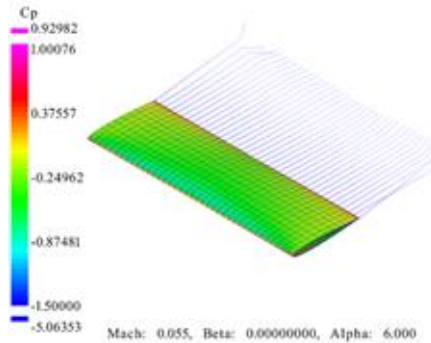
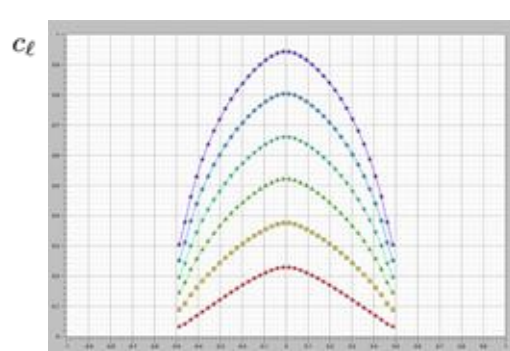
Avionics Packaging



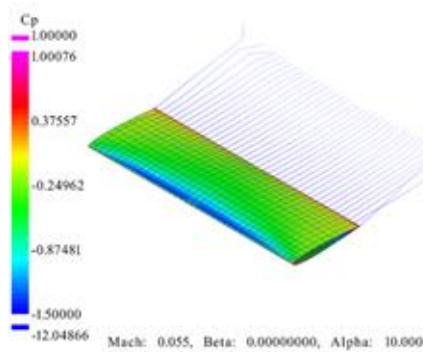
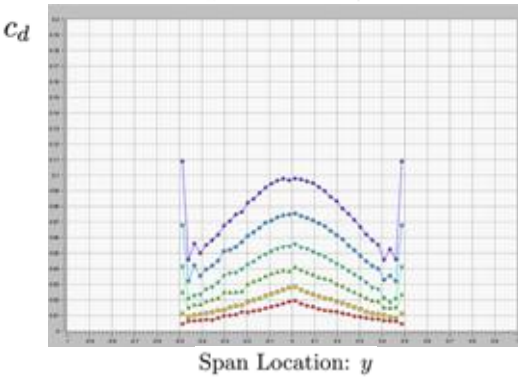
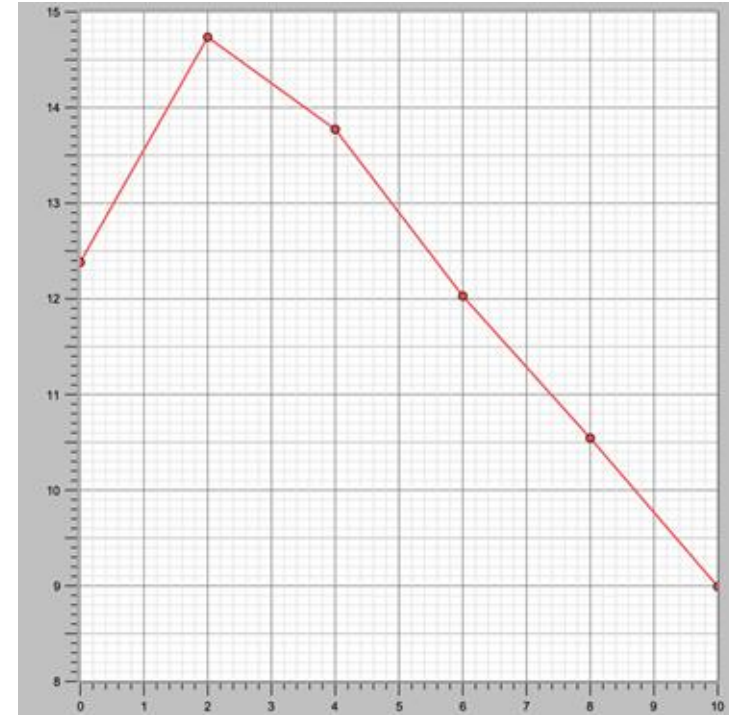
Avionics and Sensing Payload Packaging



Airfoil Selection using OpenVSP

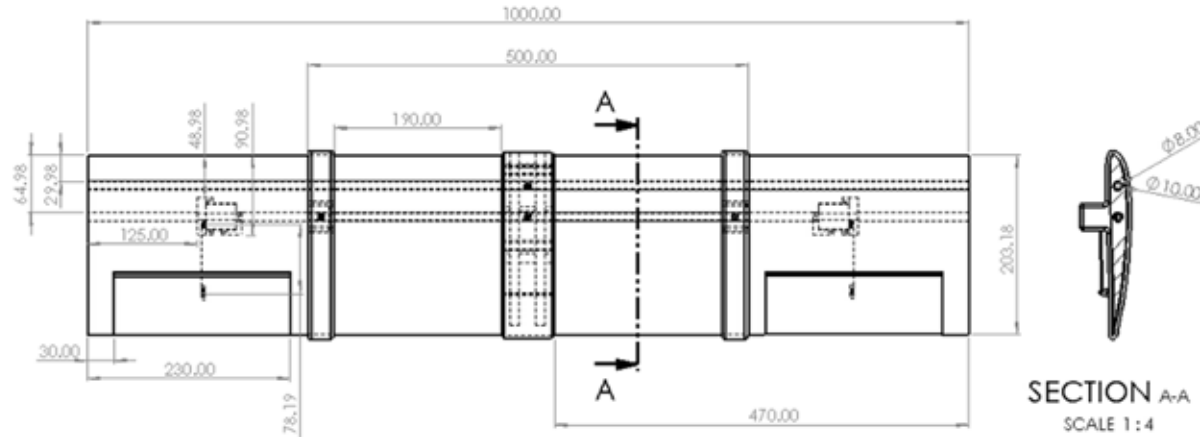


L/D

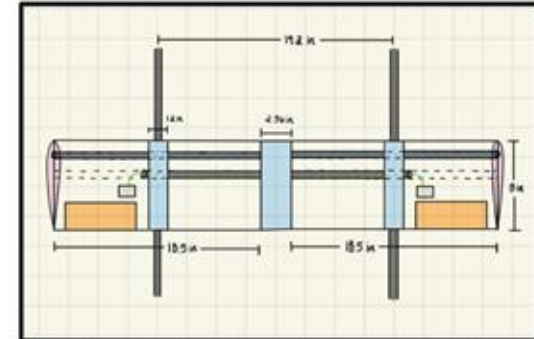


α

Wing Design



- **Span/layout:** 1000 mm span; section A–A checks fit/clearance
- **Structure:** main spar for bending; aft spar for torsion
- **Ailerons:** hinge/planform set for roll authority
- **Servos:** 125 mm inboard of wing tip; 51 × 42 mm bay
- **Wiring:** pass-through just forward of aft spar
- **RF:** tip antennas for max separation



Wing Spar Sizing

Carbon fiber strip: what width and height?

What Young's modulus for carbon fiber should I use?

$$E = 140 \text{ GPa}$$

Simply supported 48" wing with central load of 35 N, the peak bending moment at the center is:

$$M = \frac{PL}{4} = \frac{(35 \text{ N})(1.2192 \text{ m})}{4} = 10.668 \text{ N} \cdot \text{m}$$

Section Modulus (S): geometric property of cross section that links bending moment to the peak bending stress:

$$\sigma_{\max} = \frac{M}{S}, \quad S_{\text{rect}} = \frac{bh^2}{6}$$

$$\sigma_{\text{allow}} = 250 \text{ MPa}$$

$$S = \frac{10.668 \text{ N} \cdot \text{m}}{250 \text{ MPa}} = 4.2672 \times 10^{-8} \text{ m}^3$$

Distance between caps: $d \approx 25 \text{ mm}$

$$S = Ad$$

(where A is the area of one cap.)

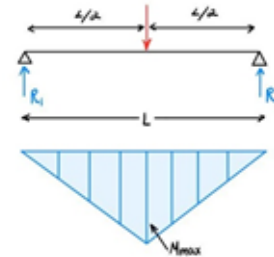
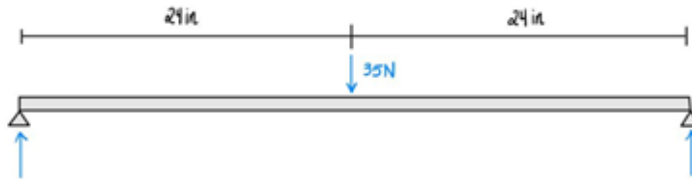
$$\frac{S}{d} = A = 1.7 \times 10^{-6} \text{ m}^2$$

$$A = 1.7 \text{ mm}^2$$

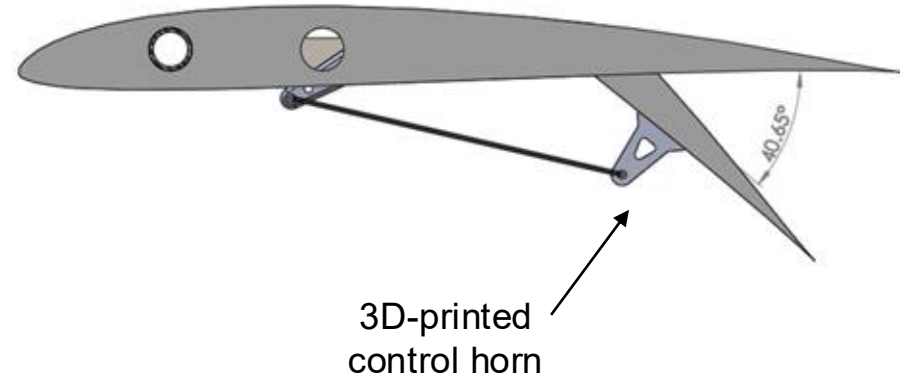
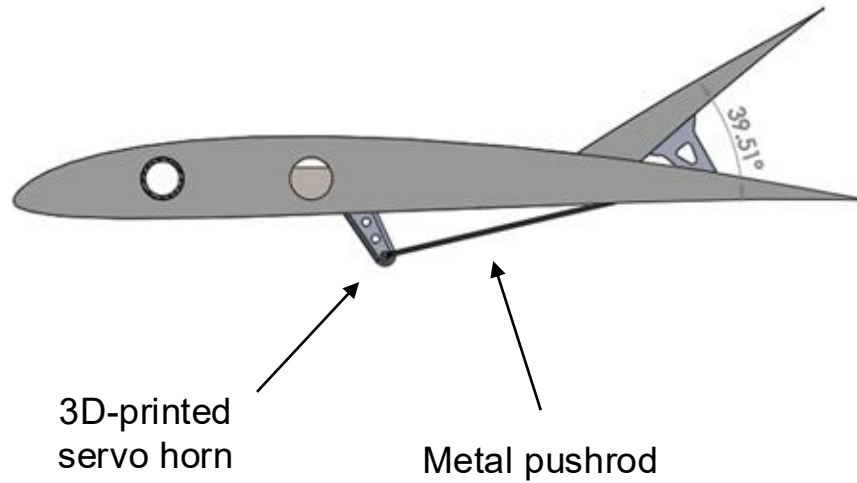
3 mm × 0.6 mm carbon fiber strip

Incorporate greater margin of safety:

$$3 \text{ mm} \times 1 \text{ mm} = 3 \text{ mm}^2$$



Aileron 4-bar Linkage



Stability Testing – Shifting the CG



Static Stability Assessment (CAD → AVL)

Method: SolidWorks mass properties and CG evaluation → AVL stability derivatives

AVL condition: $V = 22$ m/s, trimmed at $\alpha = 5^\circ$

Stability criterion: stable requires $C_{m\alpha} < 0$ (concerning result: $C_{m\alpha} > 0$)

Baseline (AVL moment reference):

$$X_{CG} = 0.1660 \text{ m}, \quad X_{NP} = 0.1175 \text{ m}, \quad \bar{c} = 0.2032 \text{ m}$$

$$SM = \frac{X_{NP} - X_{CG}}{\bar{c}} = \frac{0.1175 - 0.1660}{0.2032} = -0.239 \approx -23.9\%$$

Implication: negative static margin $\Rightarrow$ unstable; move X_{CG} forward and/or shift X_{NP} aft.

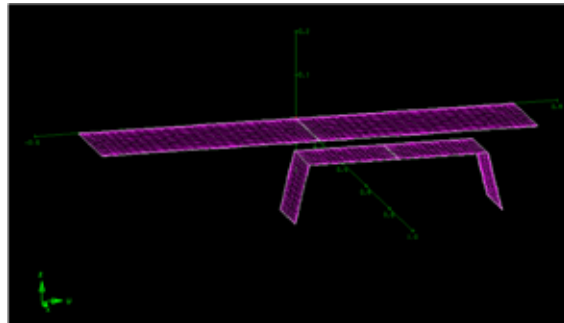
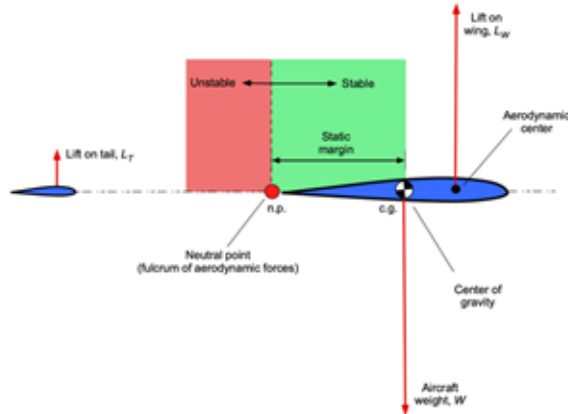
CG shift attempt (AVL case):

$$\Delta x = 3 \text{ in} = 0.0762 \text{ m}, \quad X_{CG, \text{new}} = 0.12613 \text{ m}$$

$$SM_{\text{new}} = \frac{0.117298 - 0.12613}{0.20320} = -0.0435 \approx -4.35\%$$

Result: CG shift improves SM magnitude but remains negative (still unstable).

Stability Testing – Shifting the Neutral Point



Neutral Point Shift via Tail Sizing

Goal: shift X_{NP} aft to achieve positive static margin (target $SM_t = 0.10$).

$$X_{NP,t} = X_{CG} + SM_t \bar{c} \quad (X_{CG} = 0.12613 \text{ m}, \bar{c} = 0.20320 \text{ m} \Rightarrow X_{NP,t} = 0.14645 \text{ m})$$

Neutral point model (tail contribution):

$$X_{NP} = X_{ac,ref} + \Delta X_{NP,t}$$

Sizing assumption:

$$X_{NP}(S_t) - X_{ac,ref} \propto S_t$$

Wing-only reference (no horizontal tail):

$$X_{ac,ref} \equiv X_{NP,wing} = 0.047602 \text{ m}$$

Required tail area ratio (hit $X_{NP,t}$):

$$\frac{S_{t,t}}{S_{t,0}} \approx \frac{X_{NP,t} - X_{NP,wing}}{X_{NP,0} - X_{NP,wing}} = \frac{0.14645 - 0.047602}{0.117298 - 0.047602} = 1.4185 \Rightarrow \Delta S_t \approx 41.9\%$$

Preserve aspect ratio (uniform scale):

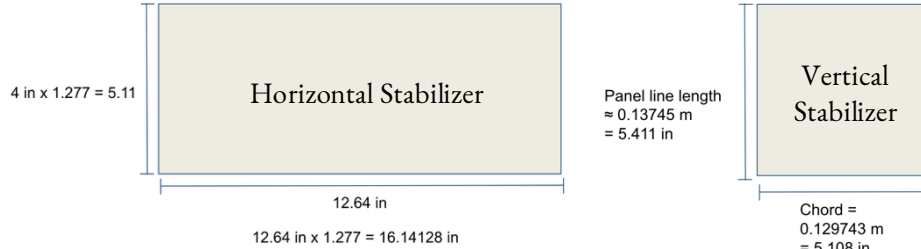
$$s = \sqrt{\frac{S_{t,t}}{S_{t,0}}} = \sqrt{1.4185} = 1.191 \Rightarrow b_t \rightarrow s b_t, c_t \rightarrow s c_t$$

AVL iteration check (same condition, $X_{CG} = 0.12613 \text{ m}$):

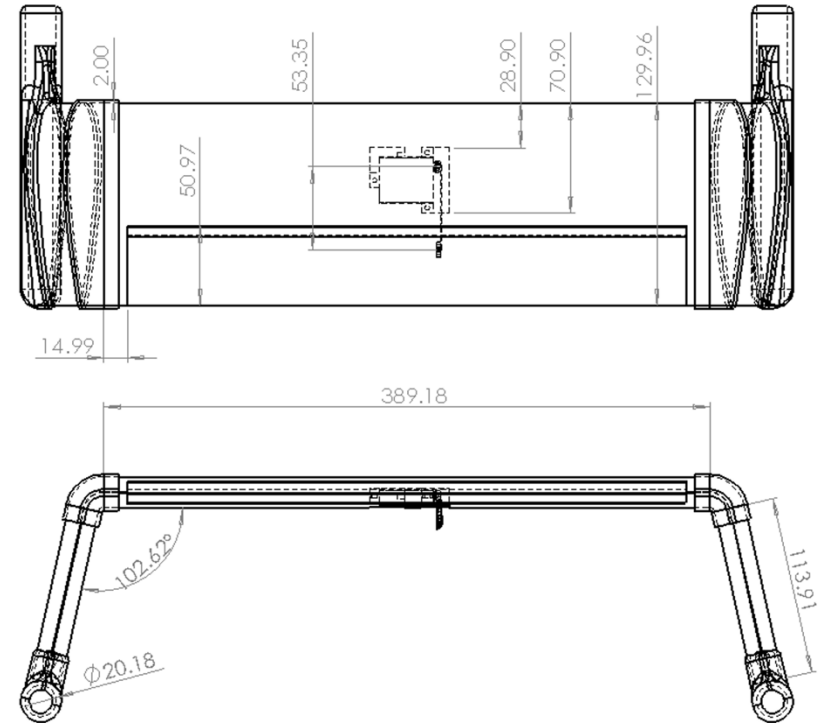
Case	s	$X_{NP} \text{ (m)}$	$SM \text{ (%)}$
Baseline	1.000	0.117298	-4.35
Iter 1	1.191	0.133684	3.72
Iter 2	1.277	0.142436	8.03

$$\Delta X_{NP,rem} = 0.14645 - 0.142436 = 0.00401 \text{ m}$$

Empennage Design



- **Baseline sizing:** h-stab + v-stab sized for 8% static margin
- **Manufacturing constraint:** attachment curvature forced shorter stabs
- **Geometry/key dims:** h-stab = 389.18 mm; h-v stab angle = 102.62° ; v-stab = 113.91 mm; tube/feature $\varnothing 20.18$ mm.



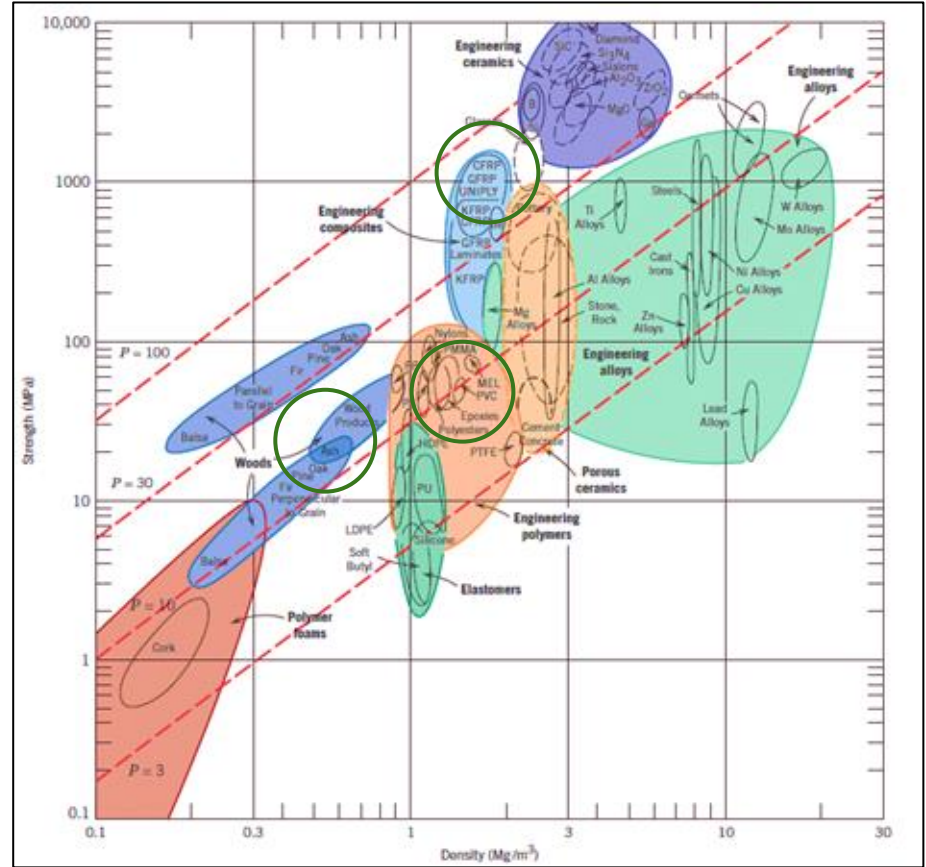
BUILD

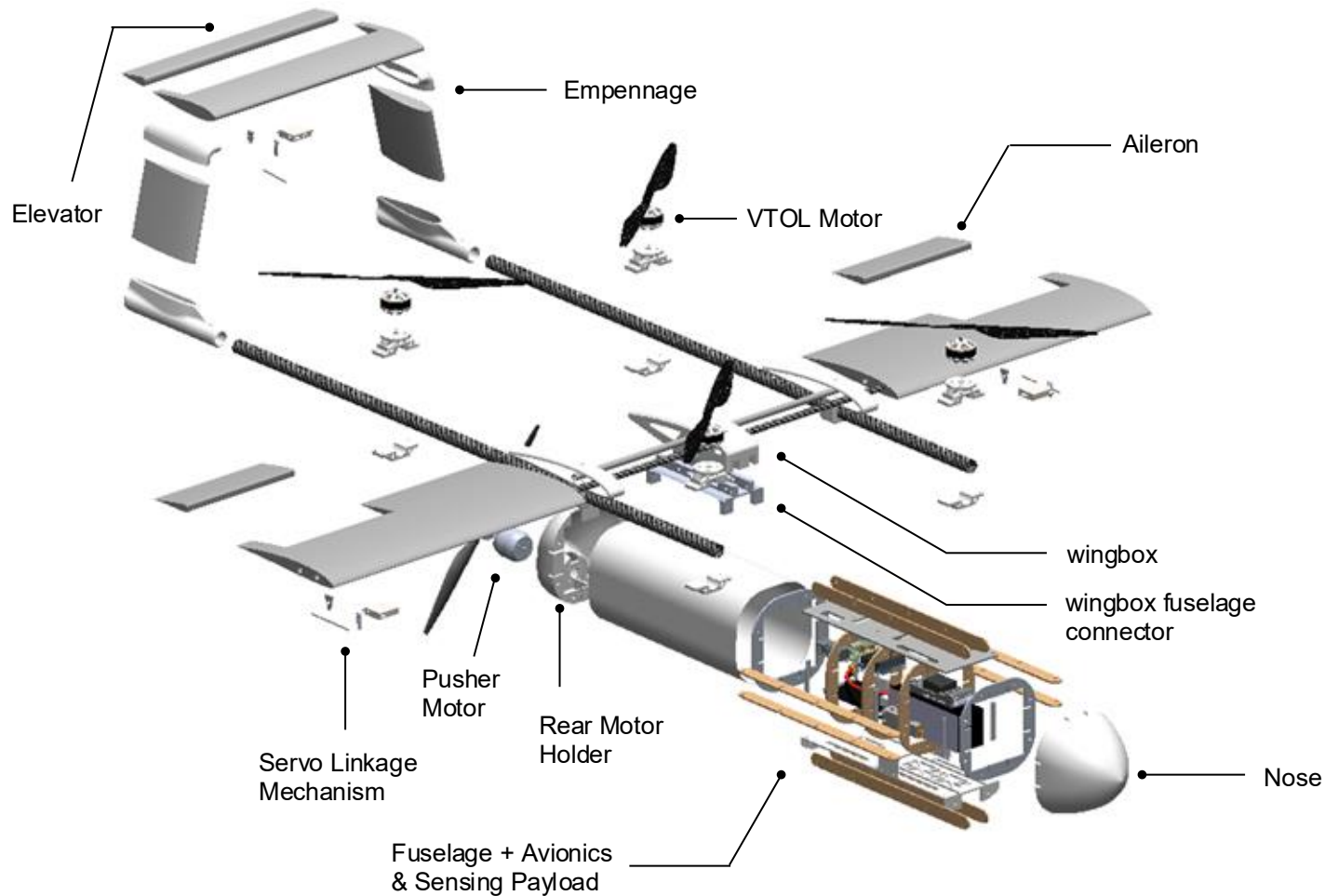


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Material Selection

Component	Material
Booms	CFRP
Wing Spars	CFRP
Wing	Foam
Nose	PLA
Rear Motor Holder	PLA
Wing Box, Boom Holders, VTOL Motor Holders	PLA
Fuselage Structure	Balsa wood



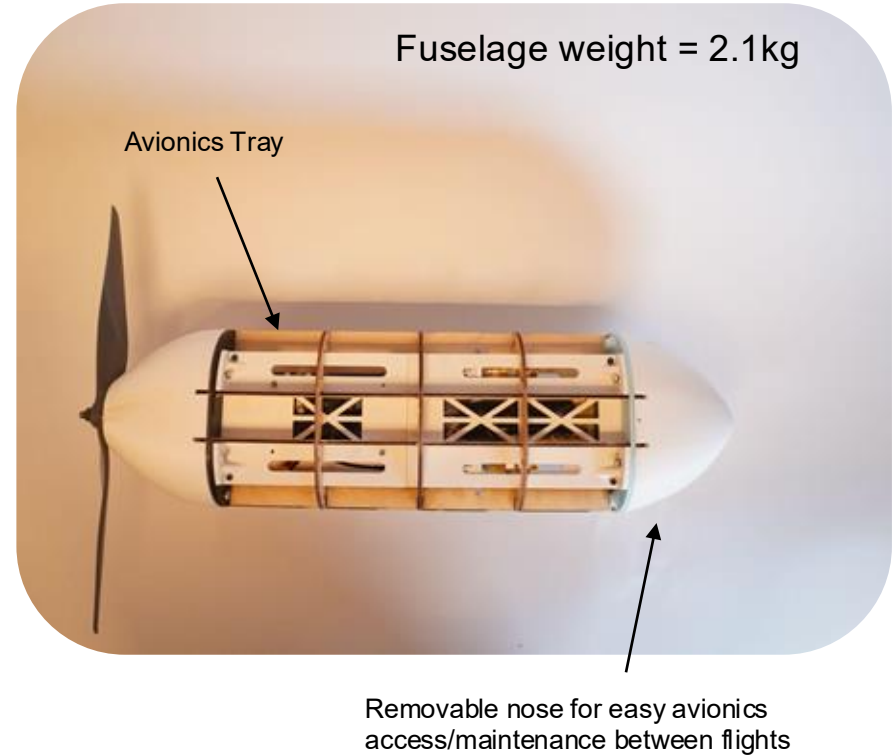
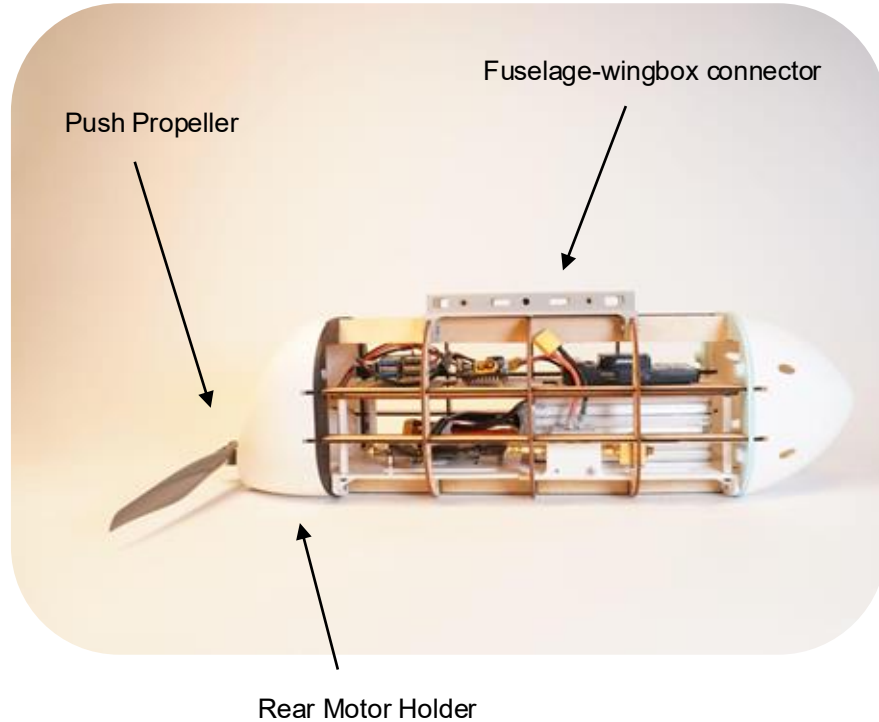


Subassembly	# parts
Empennage:	12
Wing:	17
VTOL & Booms:	18
Fuselage:	24
Nuts & Bolts:	101
Total:	172

Fuselage Build, Payload Integration, and VTOL Motor Boom Mounting

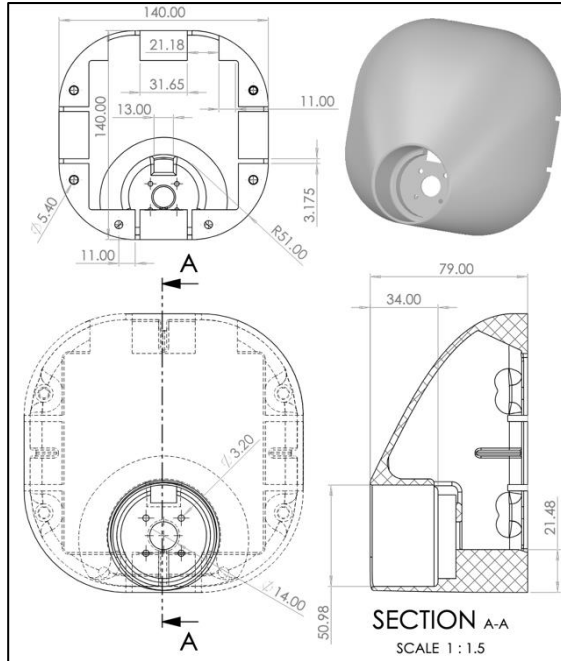


Assembled Fuselage

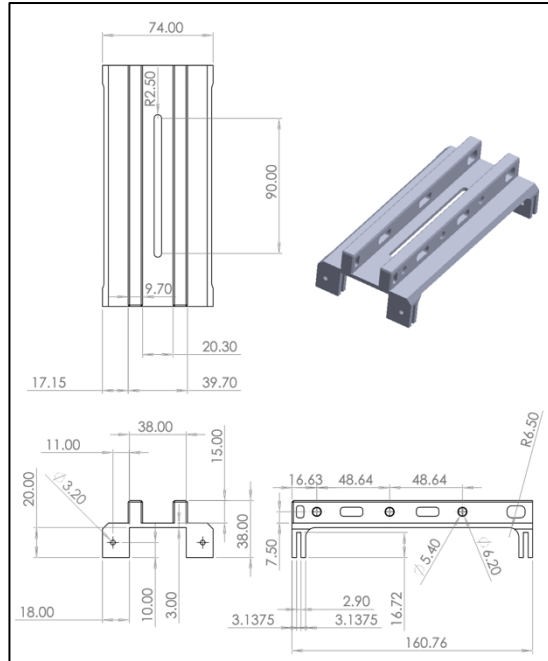


Critical 3D-printed Components

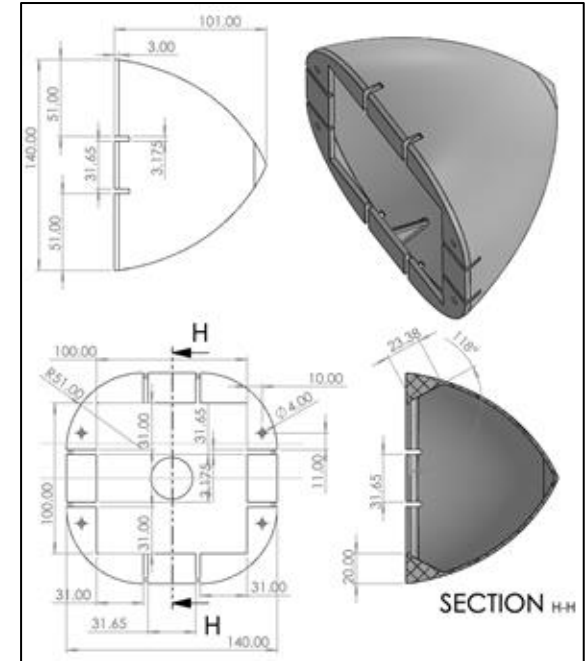
Rear Motor Holder



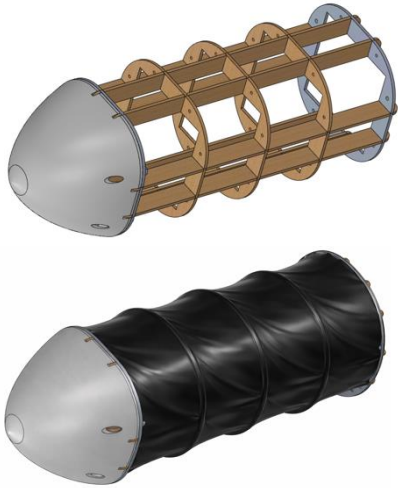
Fuselage-wingbox connector



Removable nose for easy avionics access/maintenance between flights



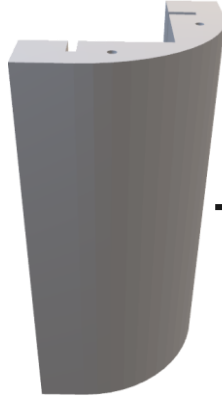
Fuselage Skin and Waterproofing Method



Initial Method: Monokote Wrap

- Iron-on film
- Rib structure causes surface scalloping

3D-printed Mold



Temporary Fixturing



Forming Skin Panel

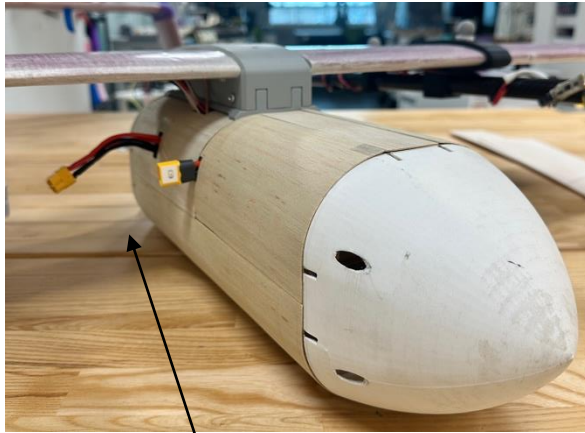


New Method: Balsa Sheeting and Fiberglass

- Water-resistant outer skin
- Balsa panels formed with ammonia + water bend technique
- 3D-printed corner-radius mold

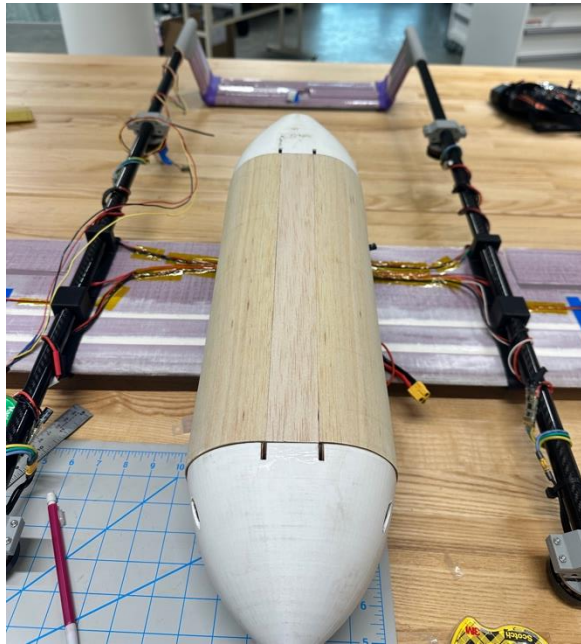
Assembled Fuselage Paneling

Right View

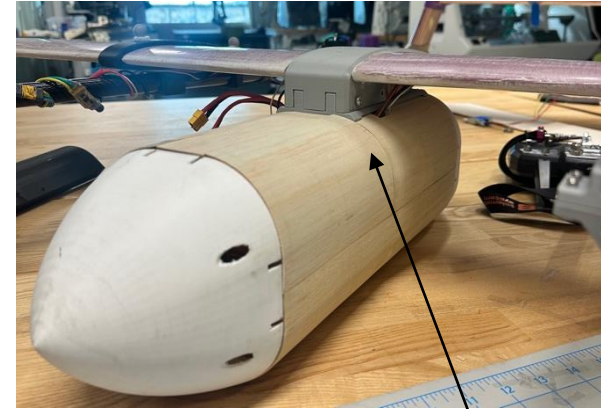


Maintained battery accessibility for operation when testing

Bottom View

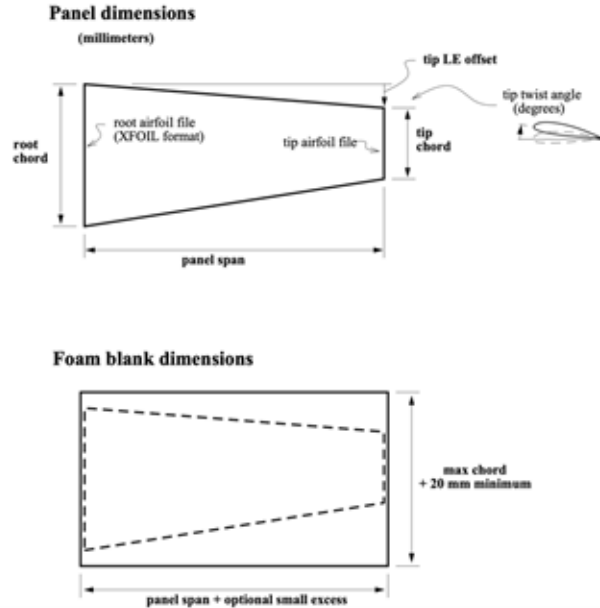


Left View



Cut meticulously to accommodate wingbox fuselage connector and motor/servo wiring

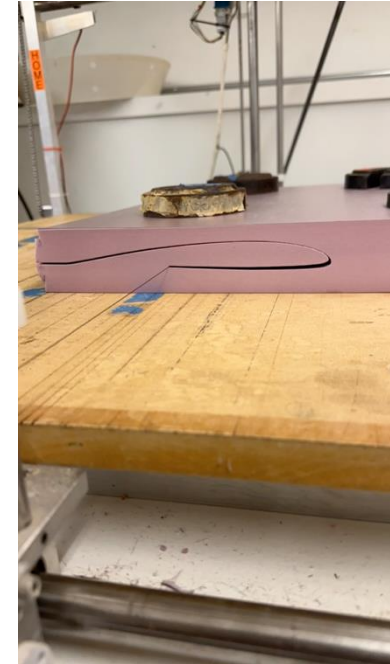
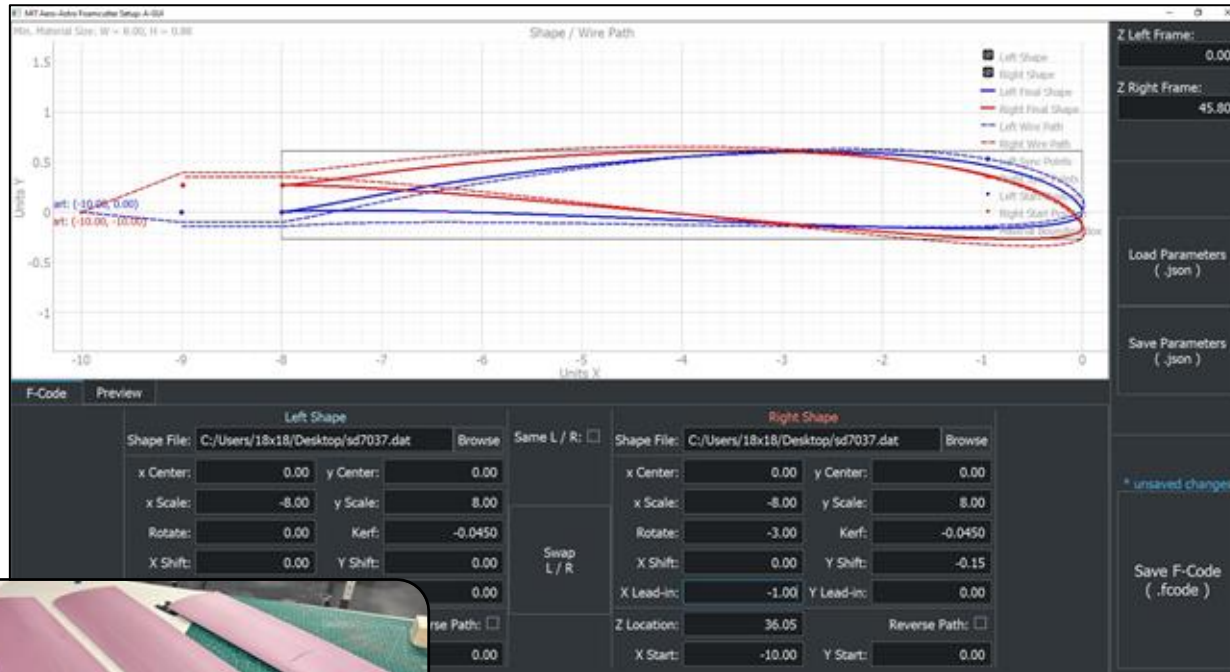
Fabrication Method



- **Hot-wire cutter:** left/right motorized carriages; current heats wire.
- **Heat + speed tradeoff:** wire voltage set by foam type; cut quality depends on temperature vs feed rate
- **2-sided motion:** independent left/right motion enables taper + twist
- **Software workflow:** import/scale .dat airfoil, add lead-in/out, generate F-code → set cut speed, execute toolpath.

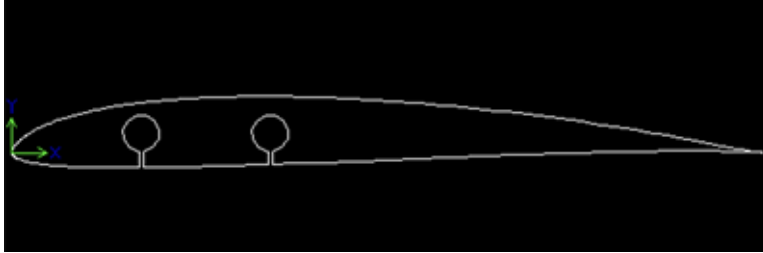
CNC Hot Wire Foam Cutter

Prototyping Techniques to Refine Wing Fabrication

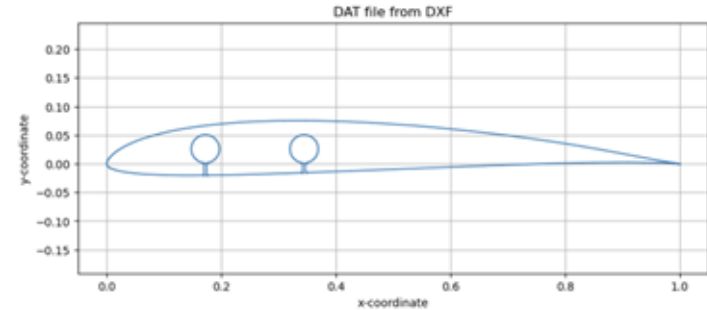


CNC Foam Cutter G-Code

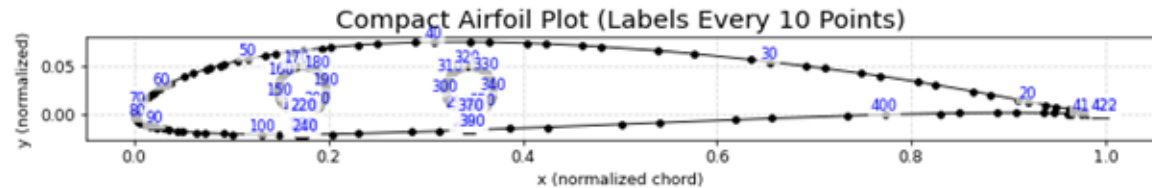
DXF



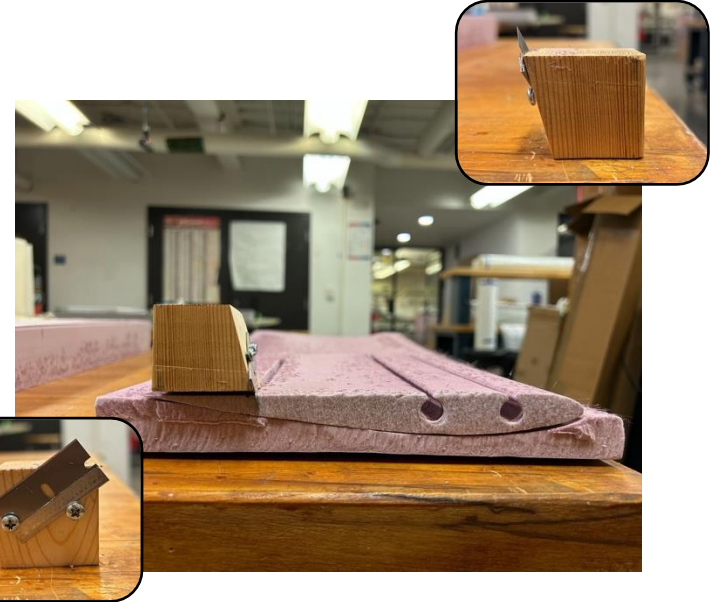
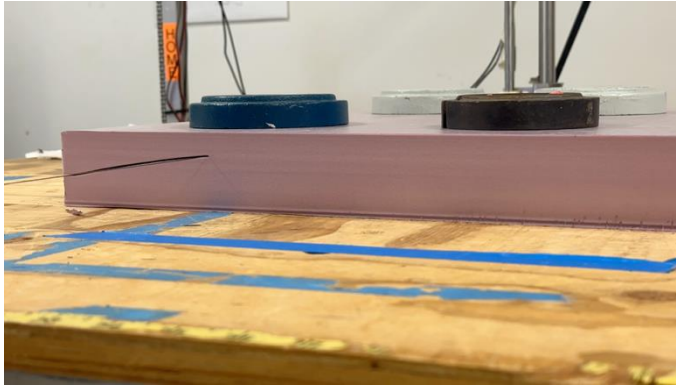
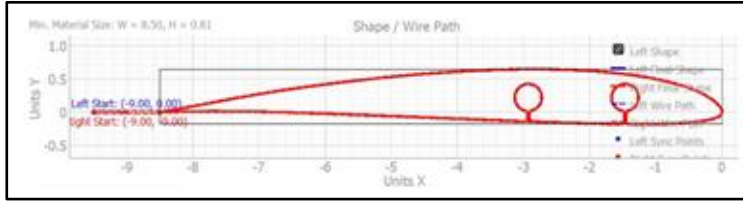
DAT(normalized chord)



Hotwire foam cutter can only cut continuous loops so we wrote a python script to convert DXF files to a DAT format (gcode with ordered points(starting from leading edge) for cutting airfoils)



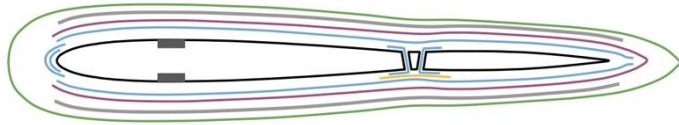
Wing F-code Execution and Aileron Construction



- Chord scaling $X = -8.50$, $Y = 8.50$ with $\text{kerf} = -0.0350$
- Fixed position with lead-in = -1.00
- Wire voltage = 16 V; wire speed = 2.2 in/min

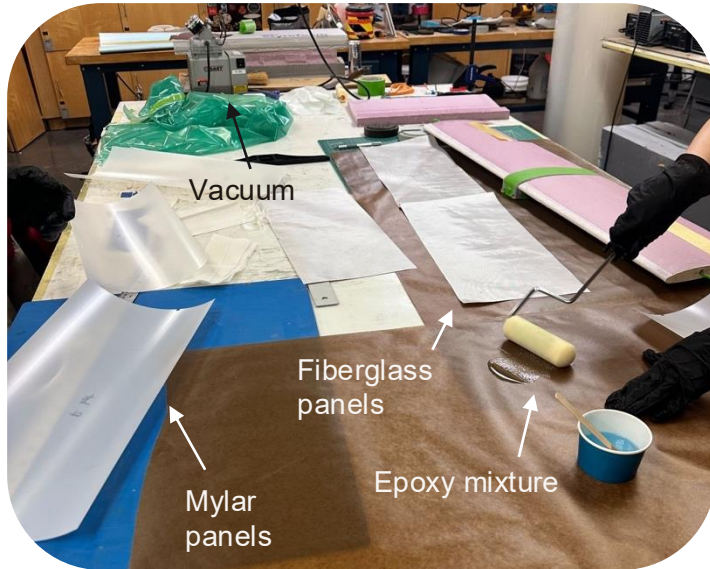
- Blade angle = 15° , aileron deflection = 30°
- Fixed for consistent hinge line and guided cut

Mylar-bagged Fiberglass Layup



- Vacuum Bagging
- Breather
- Mylar
- 1.7 oz Kevlar
- 1.5 oz Fiberglass

weights to ensure proper fit before vacuum bag provides sufficient force to remove

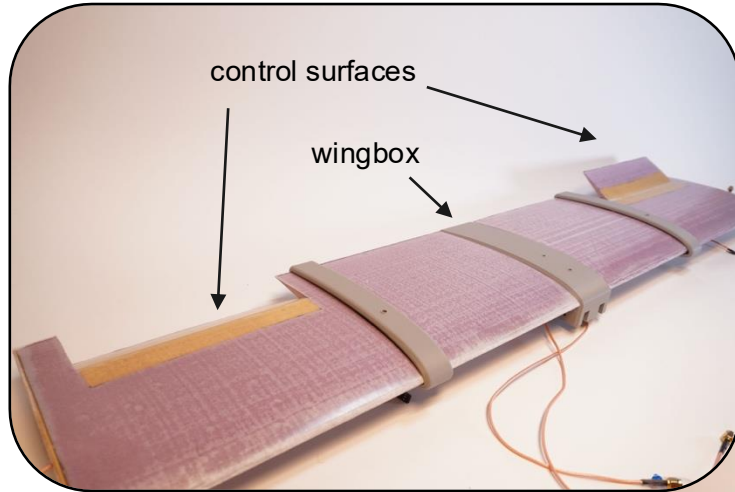


Vacuum tape

Breather cloth



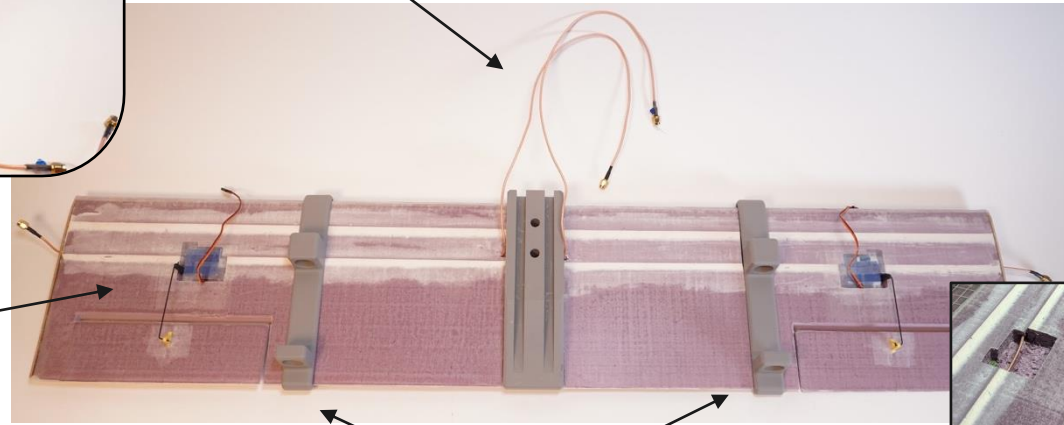
Final Wing Structure



Control surface assembly

SMA cable routed through wings to enable 1m antenna separation

servo mechanism

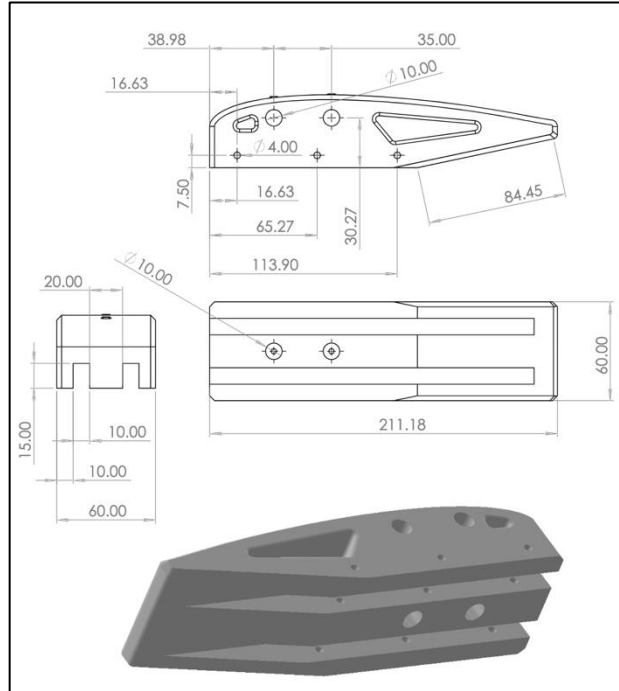


boom holders

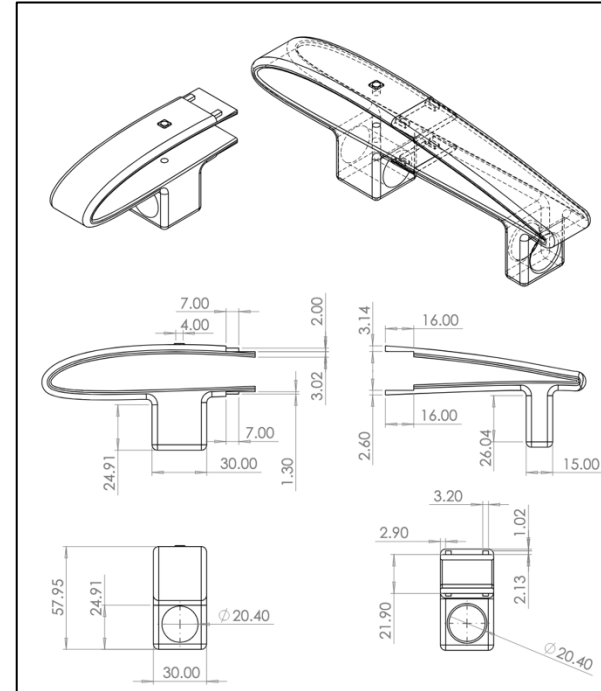


Critical 3D-printed Components

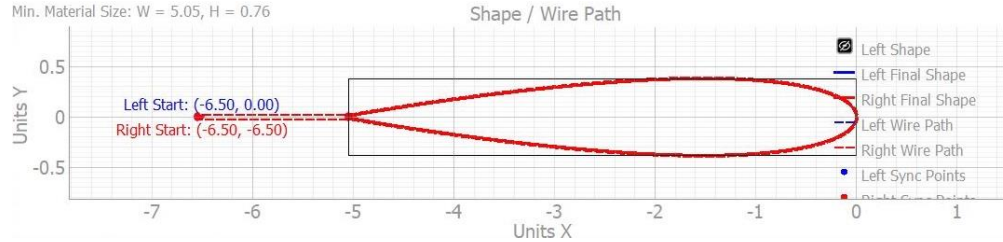
wingbox



boom holders

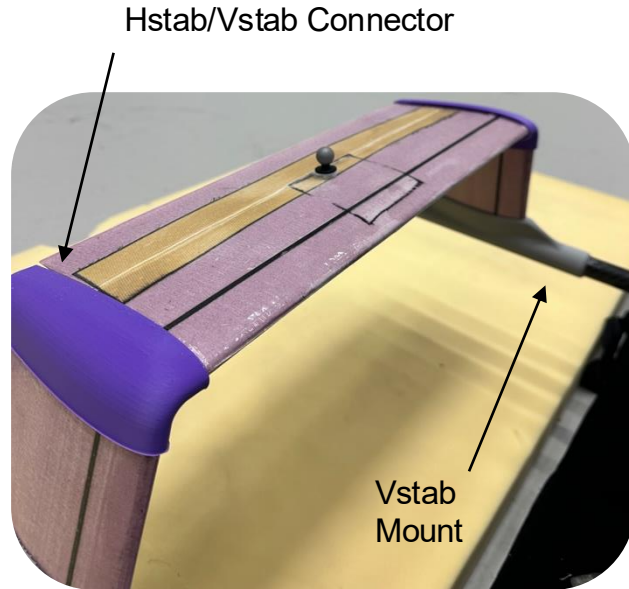


Empennage Fabrication

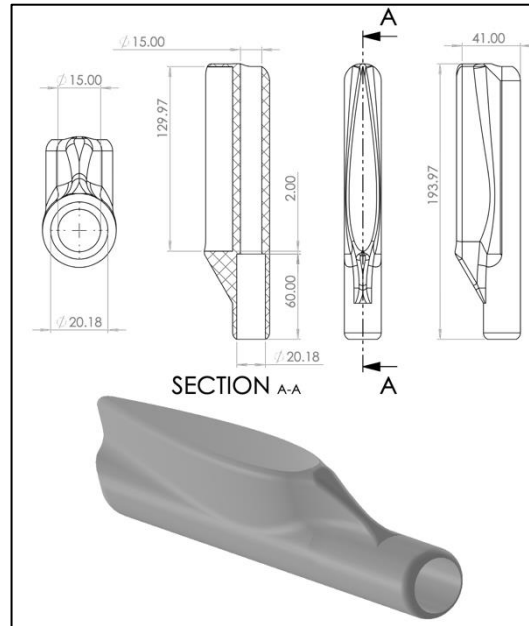


- **Foam core fabrication:** Horizontal and vertical stabilizers are hot-wire cut.
- **Spar integration:** Spar channels are 1 in aft of the leading edge and 1 mm deep; placement preserves clearance
- **Composite finishing:** Mylar-bagged fiberglass layup

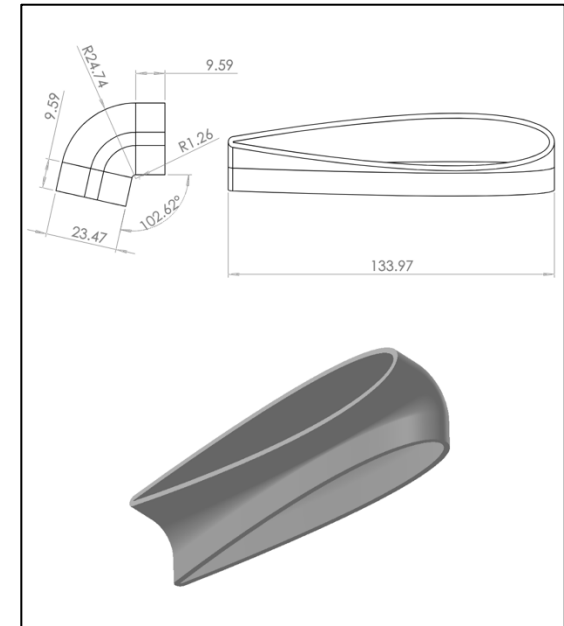
Empennage Assembly and 3D-printed Connectors

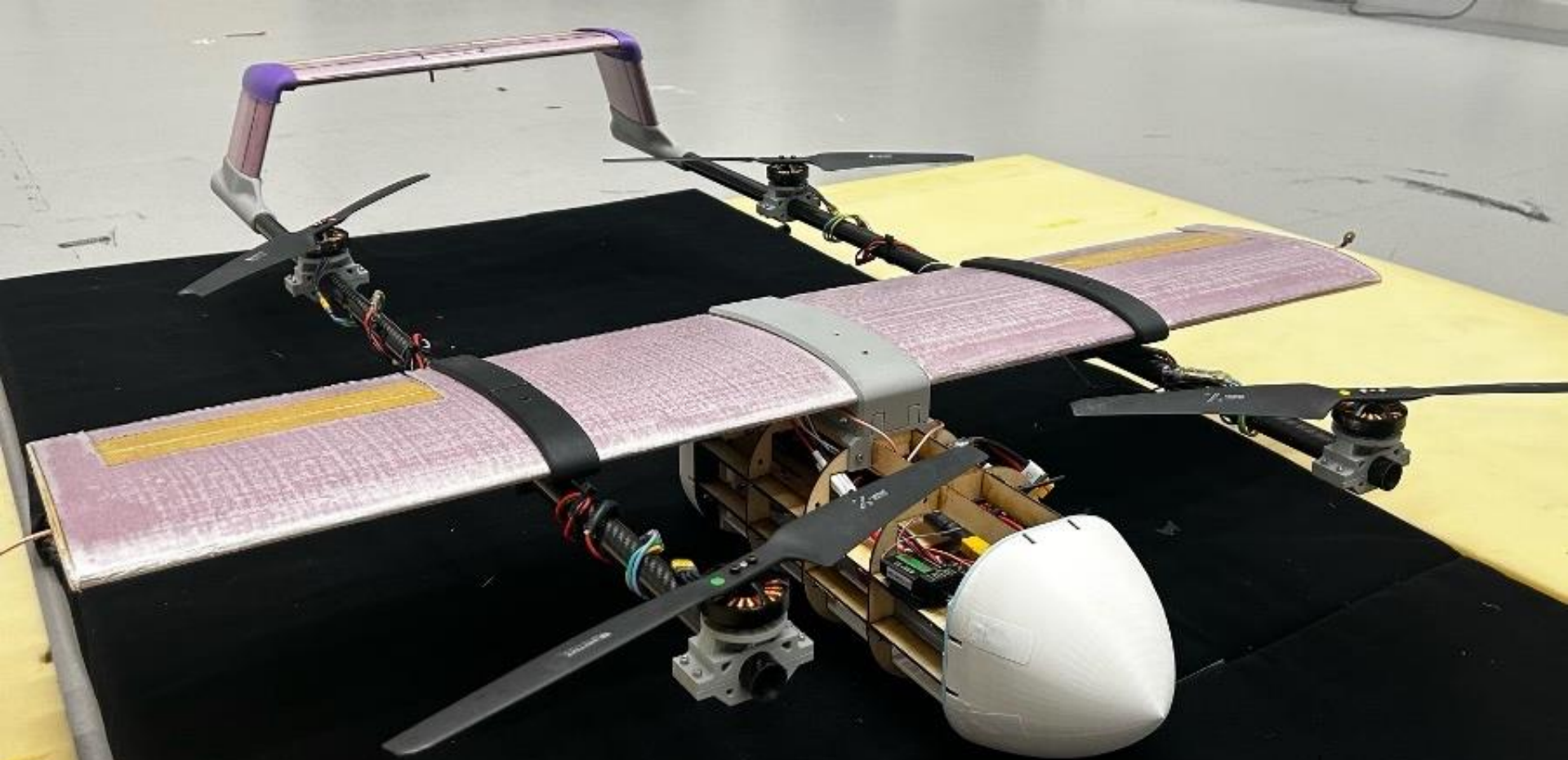


Vstab Mount



Hstab/Vstab Connector





TEST-READY PROTOTYPE



Harvard John A. Paulson
School of Engineering
and Applied Sciences

BUILD



MEASURE & VERIFY



Harvard John A. Paulson
School of Engineering
and Applied Sciences

Center of Gravity Test and Static Margin Verification



Neutral Point, X_{np} : 0.142436 m

Measured CG, X_{cg} : 0.11100 m

Mean Aerodynamic Chord, c : 0.2032 m

Static Margin:

$$SM = \frac{X_{np} - X_{cg}}{c} = 15.47\% \text{ MAC}$$

Predicted CG: 0.12613 m

Measured CG: 0.11100 m

QGroundControl Flight Configuration



Actuators Config

Geometry: Standard VTOL

Motors 5

Position X: Position Y: Direction CCW Axis

Motor	Position X	Position Y	Direction CCW Axis
Motor 1	0.30	0.24	☒
Motor 2	-0.35	-0.24	☒
Motor 3	0.30	-0.24	☐
Motor 4	-0.35	0.24	☐
Motor 5	-0.07	0.00	☐

Control Surfaces 3

Type	Roll Scale	Pitch Scale	Trim
Servo 1: Left Aileron	-0.50	0.00	
Servo 2: Right Aileron	0.50	0.00	
Servo 3: Elevator	1.00	0.00	

Lock control surfaces in hover: Enabled

Motor Position: relative to the measured CG, CW and CCW distinction

Actuator Outputs

PWM AUX PWM MAIN UAVCAN

MAIN 1-2 PWM 400 Hz

Function	Disarmed	Minimum	Maximum	Rev Range (for Servos)
MAIN 1: Motor 1	1000	1100	1900	☐
MAIN 2: Motor 2	1000	1100	1900	☐

MAIN 3-4 PWM 400 Hz

Function	Disarmed	Minimum	Maximum	Rev Range (for Servos)
MAIN 3: Motor 3	1000	1100	1900	☐
MAIN 4: Motor 4	1000	1100	1900	☐

Control Surfaces: set with minimum and maximum positions of the actuated servo arms for ailerons and elevator

Actuator Outputs

PWM AUX PWM MAIN UAVCAN

AUX 1-4 PWM 50 Hz

Function	Disarmed	Minimum	Maximum	Rev Range (for Servos)
AUX 1: Disabled	1000	1100	1900	☐
AUX 2: Right Aileron	1500	800	2000	☒
AUX 3: Left Aileron	1500	800	2000	☐
AUX 4: Elevator	1500	800	2000	☒

System Integration Test

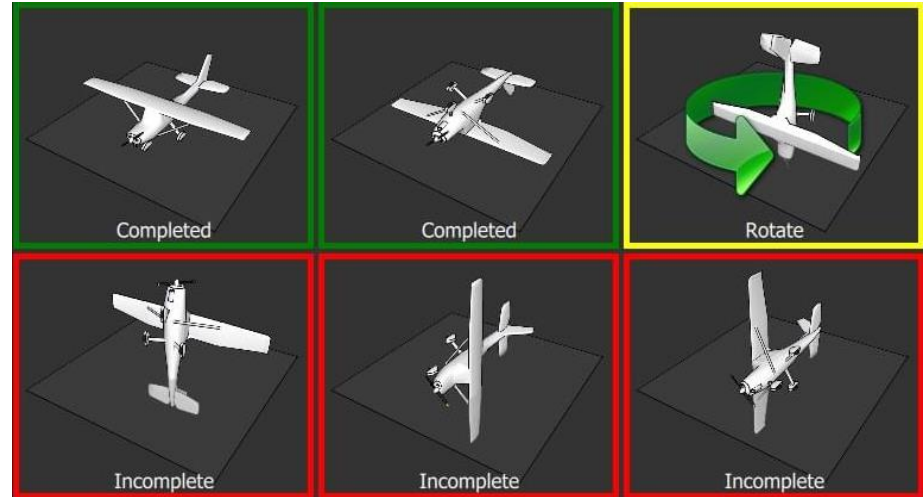
Fixed Wing Mode

VTOL Mode



Calibration for Manual-Mode Hover Test

IMU (Inertial Measurement Unit) Calibration



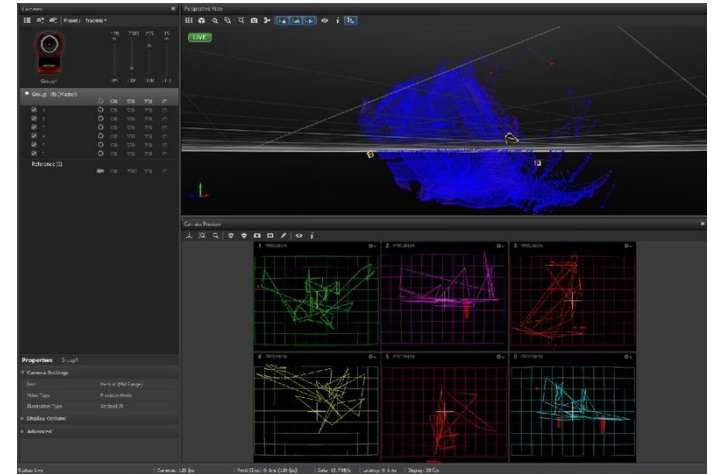
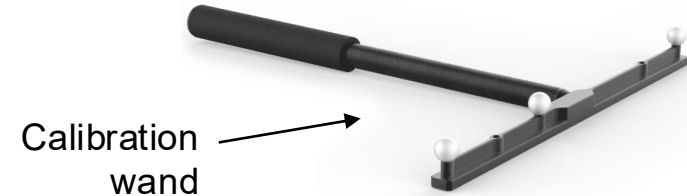
- Held drone in multiple positions to calibrate IMU
- Enables correct orientation estimation for Position mode
- Throttle controls z-axis, not motor RPM

Manual-Mode Hover Test



Necessary Calibration for Position Mode

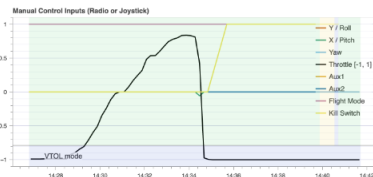
Motion Capture Camera Calibration



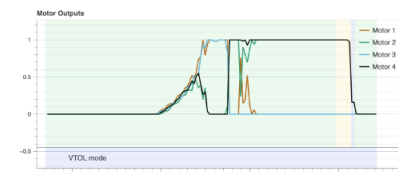
Position-Mode Hover Test I



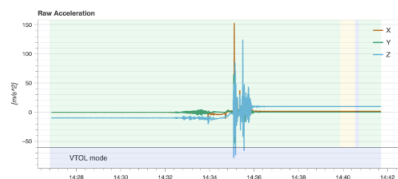
(a) Barometer Altitude



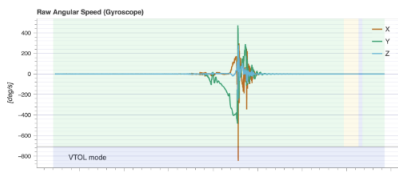
(b) Manual Control Inputs(Position Mode)



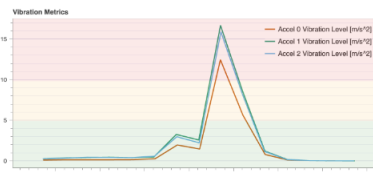
(c) Motor Output



(d) Acceleration

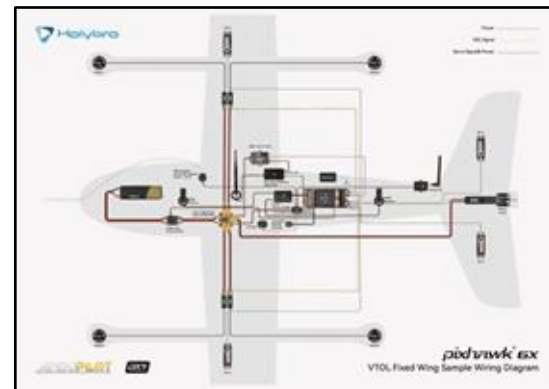


(e) Angular Speed

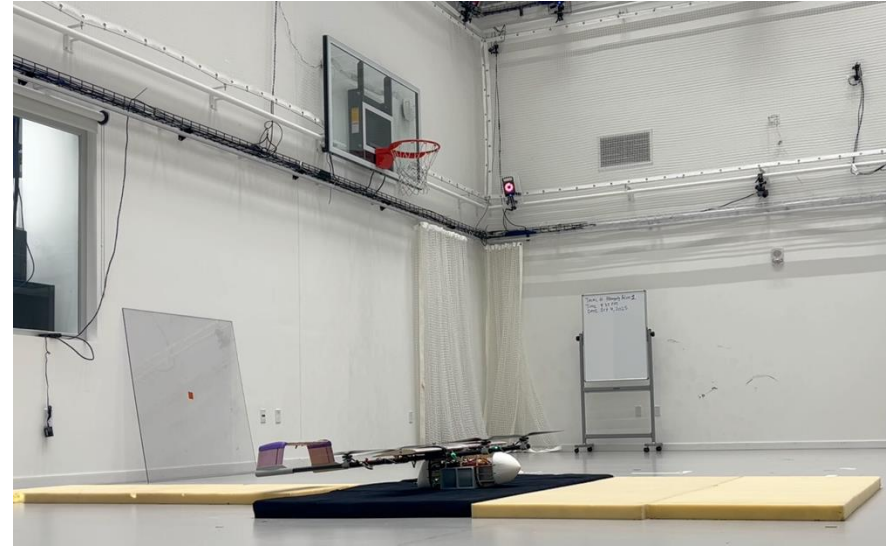


(f) Vibration

Flight Log Data From our Failed Hover test



Position-Mode Hover Test II



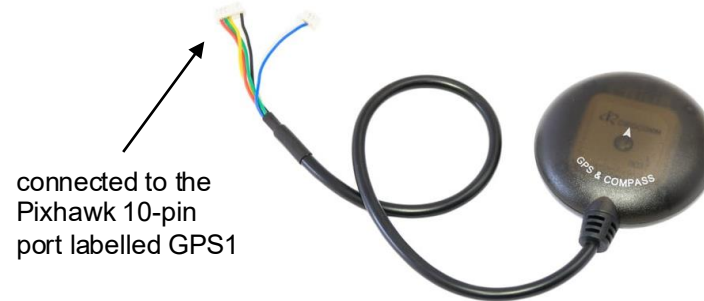
In Position mode, throttle controlled the z-axis instead of commanding equal motor RPM, allowing the flight controller to compensate for imbalances and achieve vertical takeoff and hover.

Areas for Improvement:



- Hover drift likely due to motion capture feedback
- GPS is the next step for outdoor hover and fixed-wing testing

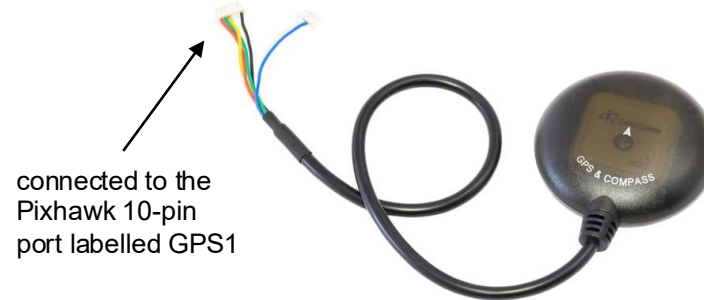
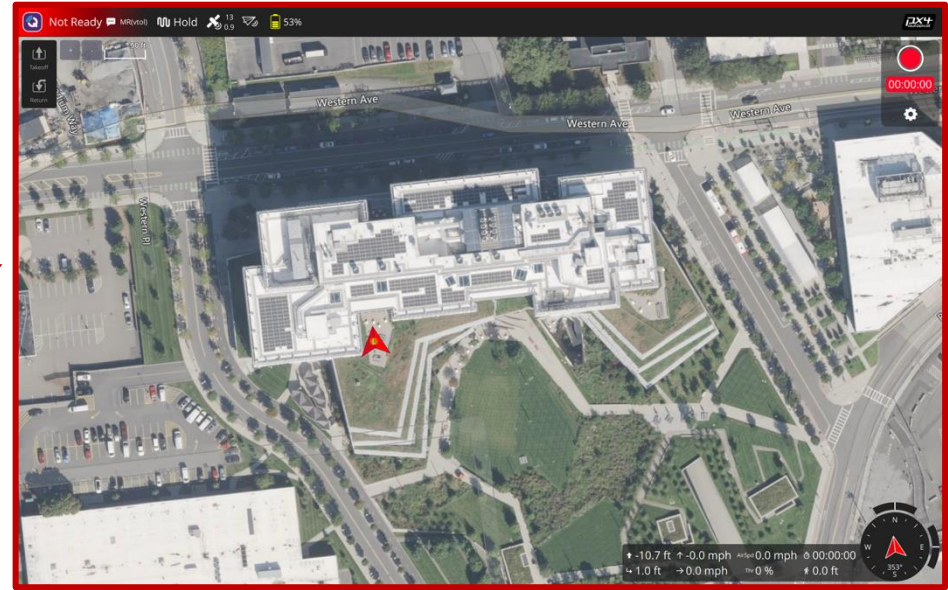
GPS Calibration



connected to the
Pixhawk 10-pin
port labelled GPS1

GNSS module:
also includes
compass/magnetom
eter, buzzer, safety
switch, and UI LED

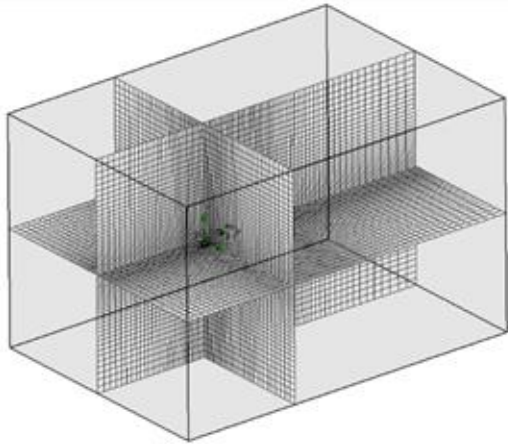
GPS Calibration



connected to the
Pixhawk 10-pin
port labelled GPS1

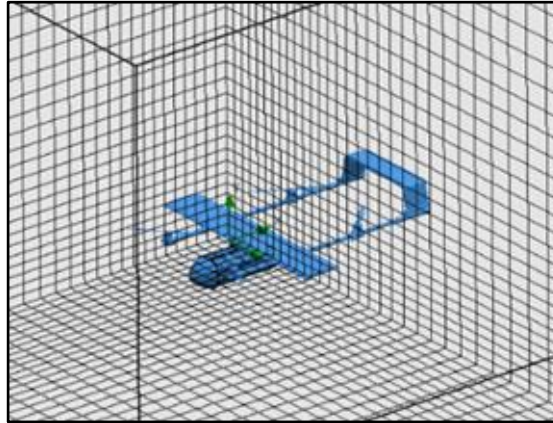
GNSS module:
also includes
compass/magnetom
eter, buzzer, safety
switch, and UI LED

CFD Setup – Refined Body & Wake Mesh



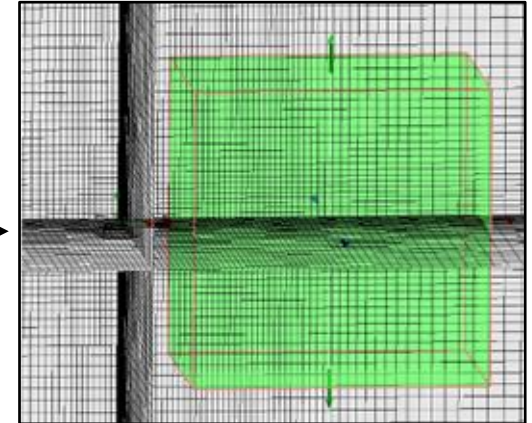
Computational Domain / Enclosure Mesh

Defines the computational fluid region and applies the base mesh for the overall flow domain.



Body Mesh Refinement

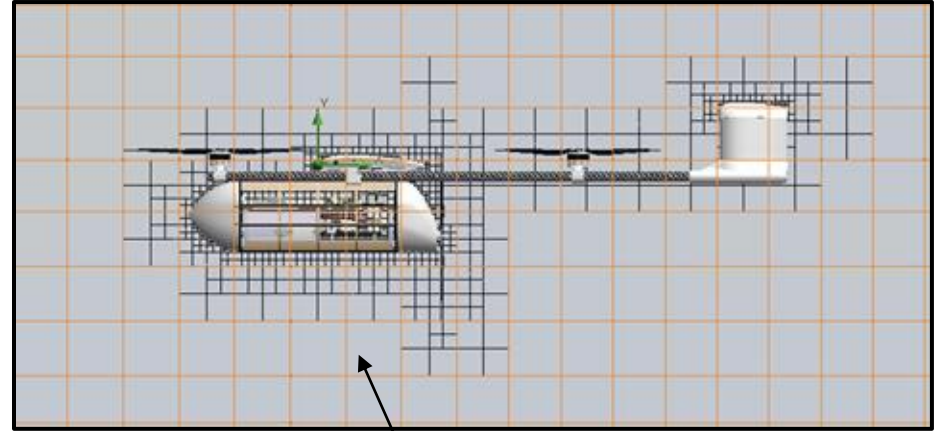
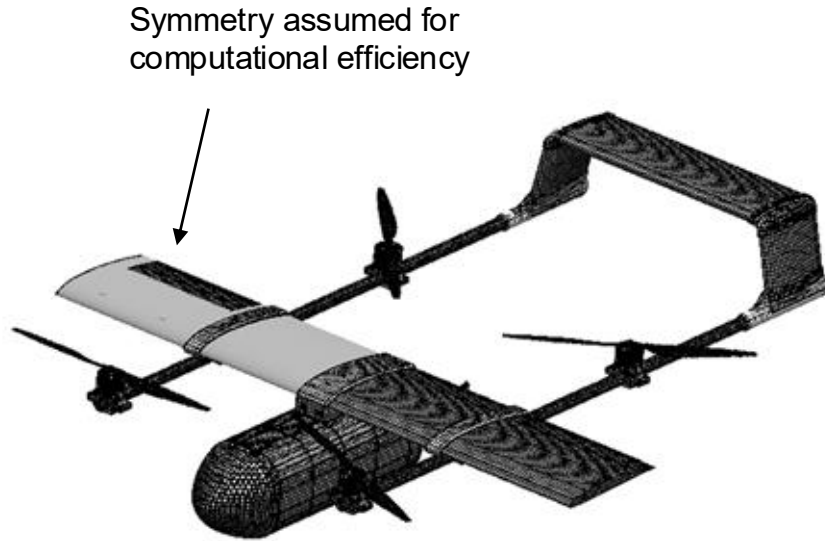
Refines the mesh around the aircraft to better capture geometry and near-body pressure/velocity gradients.



Wake Mesh Refinement

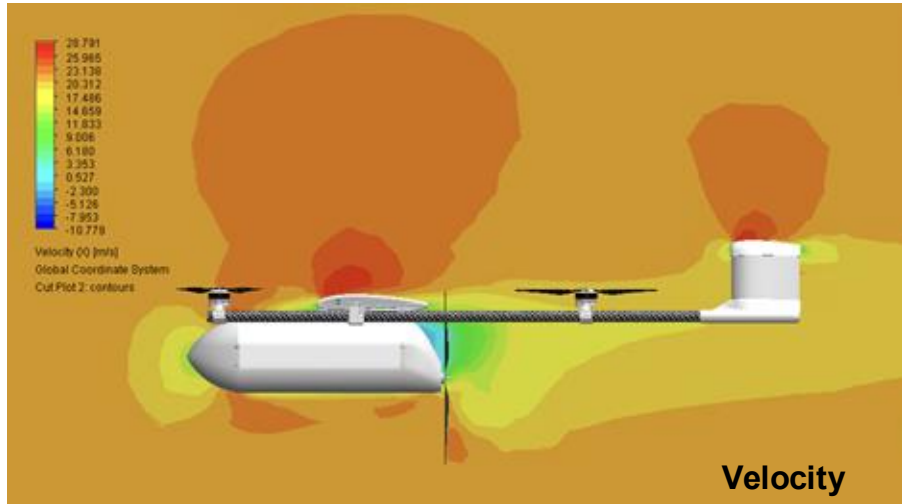
Refines the downstream wake region to better resolve separated flow and wake development.

CFD Setup – Local Refinement



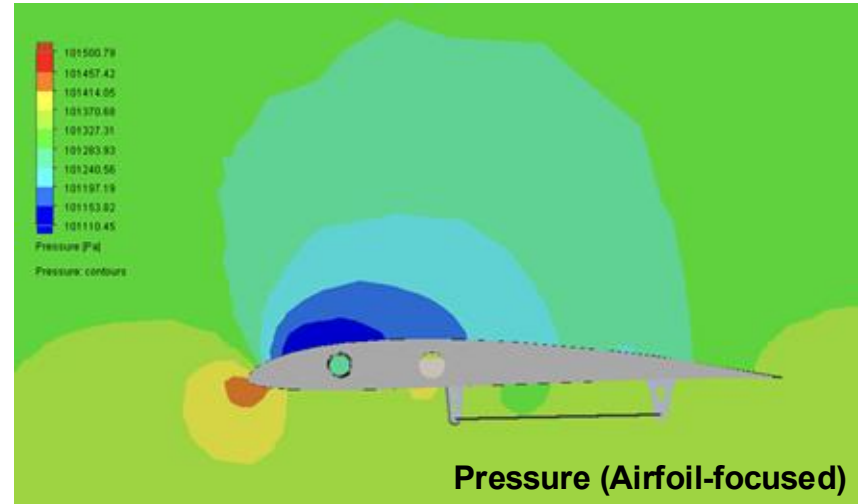
Equidistant mesh with local refinement on critical airfoils and control surfaces

Midplane Cut-plots



$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

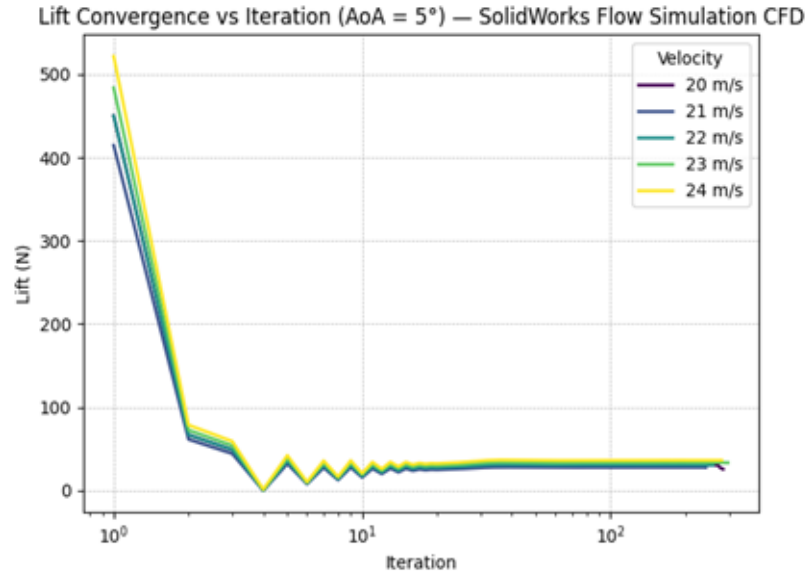
Accelerated flow over the wing corresponds to lower local pressure.



$$L = \int (p_{\text{lower}} - p_{\text{upper}}) dA$$

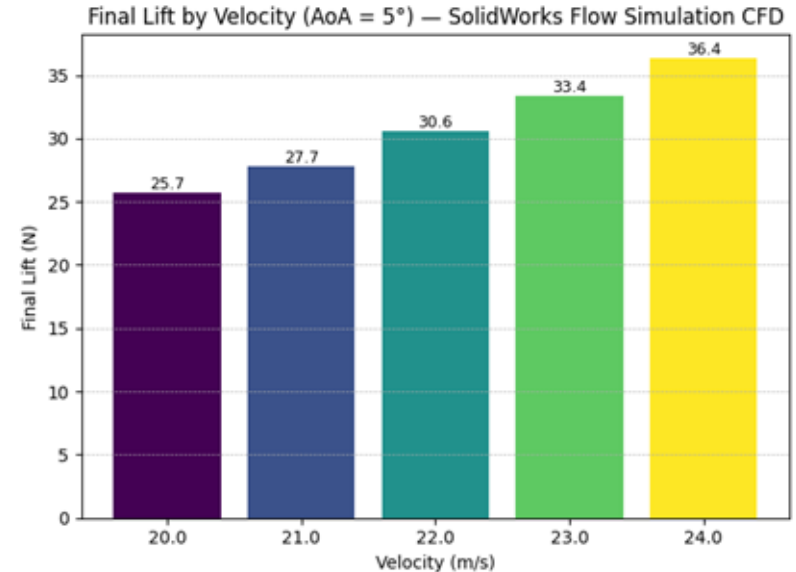
The pressure differential across the wing produces a net upward force.

Parametric CFD Study of Lift



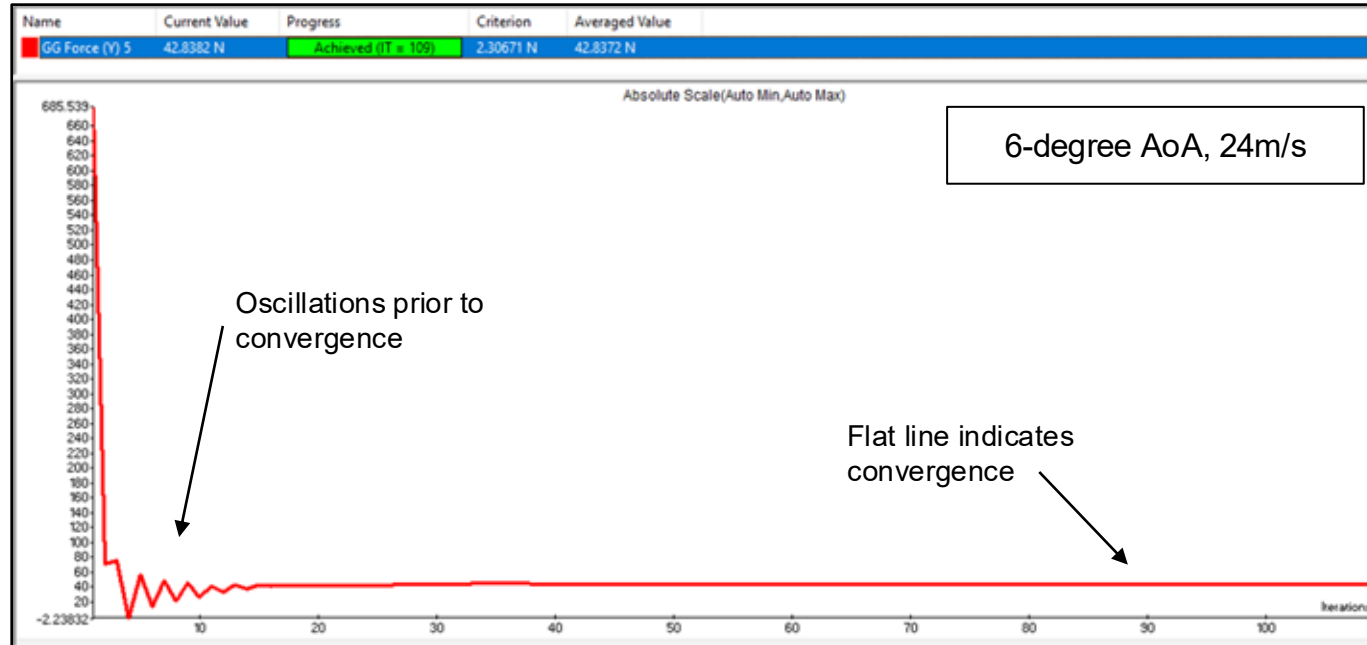
$$L = \frac{1}{2} C_L \rho S v^2$$

Force of Weight = 33.19N



Pusher Motor Specs: 24.6 m/s achieved at 90% throttle with 13-inch prop

Increased Lift Generation with AoA

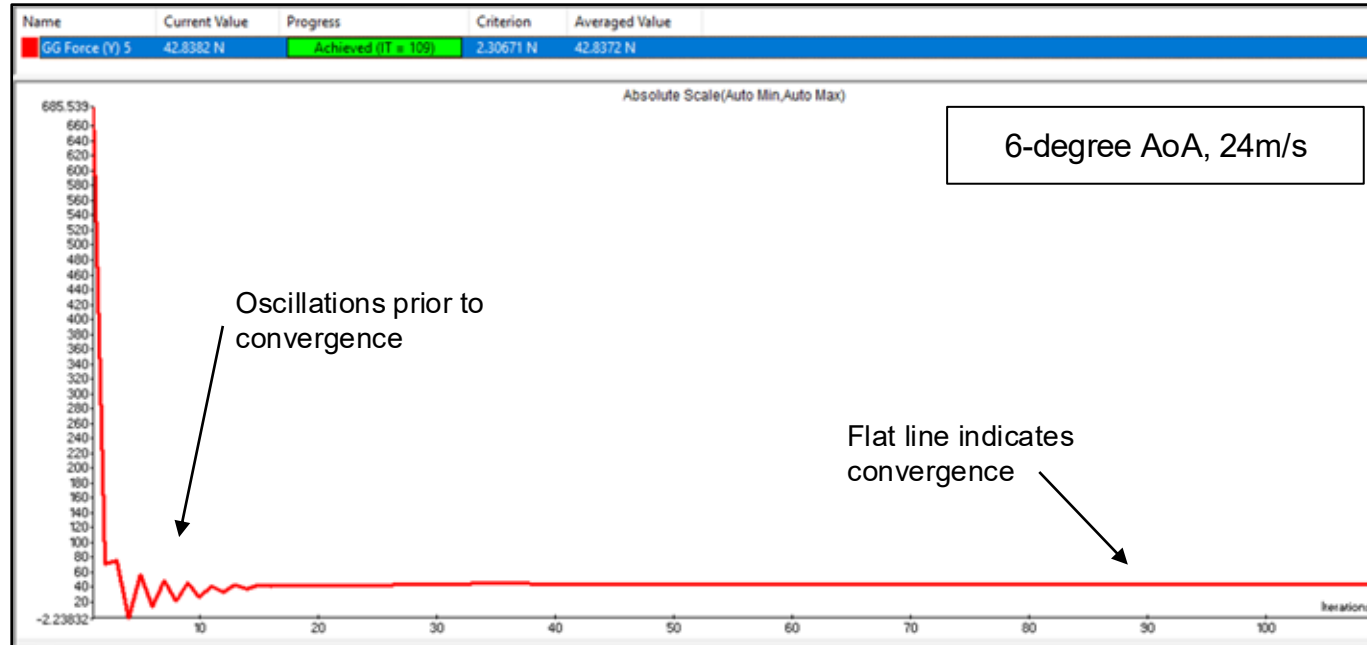


Freestream Velocity Condition

V_x	$V \cos \alpha$	23.87
V_y	$V \sin \alpha$	2.51
V_z	0	0

Increased Lift Generation with AoA

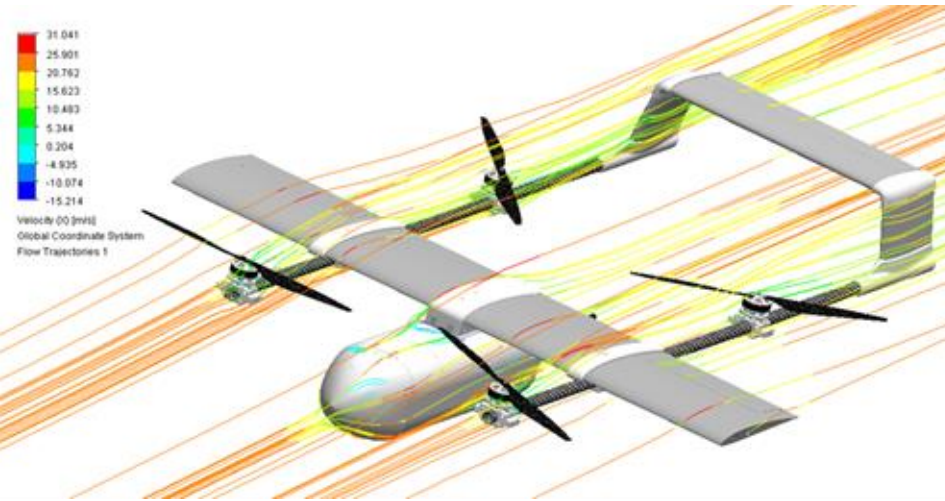
42.837 N



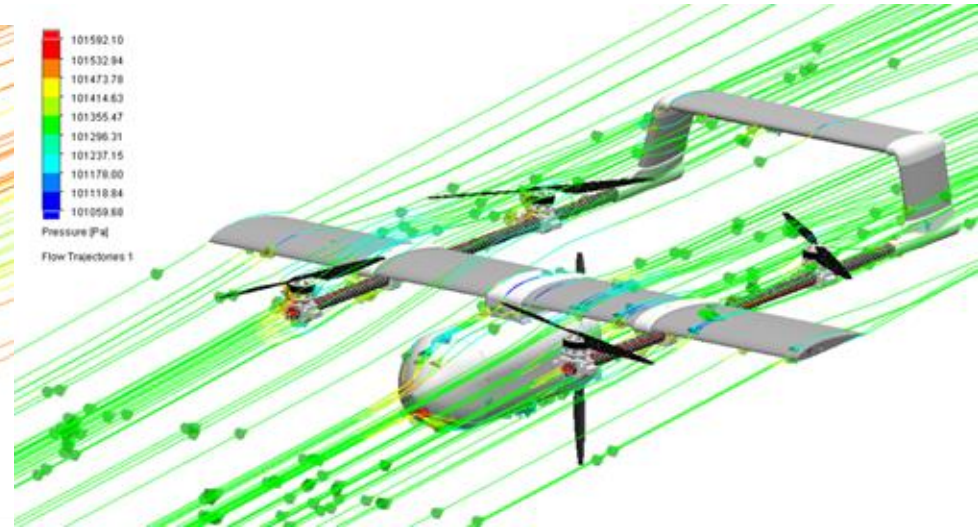
Freestream Velocity Condition

V_x	$V \cos \alpha$	23.87
V_y	$V \sin \alpha$	2.51
V_z	0	0

Flow Trajectories



Velocity



Pressure

Special Thanks to



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Aaron Vilet
David Robertson

Harvard SEAS

Seymur Hasanov
Joe Kile



Harvard John A. Paulson
School of Engineering
and Applied Sciences